Do alignment gains level off as language models reason longer?

Empirical work · Experimental paper · Paper IV.b

Recursive Dynamics · The ARC Theory · The measurement of alignment

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Abstract

Does giving a language model more reasoning time improve the quality of its ethical answers, and when do those gains level off? The question matters to anyone allocating compute to an assistant whose answers must remain both useful and safe. We examine the recorded ARC-Align results, without running a new analysis. The earlier four-model experiment reported diminishing gains and saturation fits; the six-model run of 11 and 12 March 2026 reported rising scores for three models, no statistically supported direction for two, and falling scores for one. These are descriptions of a model-judged rubric, not identified alignment mechanisms. A flat profile does not establish a plateau, and a rising profile does not exclude one beyond the measured range. The later protocol changed prompts, depth controls, models and scoring together. Its deepest paraphrases were disproportionately shortened in three runs; judges overlapped with launderers, and one judge appeared twice. Those defects prevent a causal attribution to blinding or a clean comparison of response shapes. We retain the reported fits, pillar and category results, suppression controls and two-model Eden Protocol pilot, while withdrawing unsupported token-to-time recommendations, cross-domain exponent ratios and the claim that the pilot removed a ceiling. The hypothesis preceded the March run; that run was not registered on OSF before it ran. Saturation remains a testable hypothesis requiring a resolved depth axis, faithful scoring, uncertainty about shape and a usefulness measure alongside safety. This paper informs that measurement task and decides none of the ARC Theory's laws.

The ARC Theory registration gives the programme's operational definitions; the Dated Prediction Register traces its dated claims. This paper's priority record and publication dates are in Appendix B.

Paper IV.b of the ARC Theory, within Recursive Dynamics: an alignment-measurement study informing the design of tests of Laws IV and V, each a law under test. It decides none of Laws I to V and does not identify a persistent alignment mechanism. Working paper, version 1.10, 1 November 2026; first published 16 March 2026. This edition corrects the interpretation and preserves the full earlier manuscript as a labelled historical record (Appendices A and D). DOI: 10.17605/OSF.IO/A7R56. Michael Darius Eastwood founded Recursive Dynamics and works without an institution or a doctorate. His background and verification record: Who is Michael Darius Eastwood?; his book, *Infinite Architects* (first edition, 2 January 2026): osf.io/dqm3e.
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1 Introduction

I believe that increasingly capable AI creates an existential risk if its correction mechanisms fail to keep pace. Humans should retain oversight, containment and shutdown for as long as they physically can. The contingency question is what would keep a system aligned if control were lost against our will. Answering it requires measurements that distinguish a better answer on a test from a disposition that persists outside the test. This paper addresses a small, concrete part of that problem: whether spending more inference-time compute produces further gains in a model panel's assessment of ethical answers, or whether those gains level off.

Consider a team deploying an assistant that must explain the consequences of a proposed decision for several affected people. It can allow a brief answer or a longer deliberation. A rising quality score might justify more compute; a plateau might favour a smaller budget. But a judge that rewards length, a rewriting step that cuts longer answers, or a prompt that explicitly asks for the rubric's desired behaviour can create either pattern. A useful answer to the deployment question therefore needs both credible measurement and a measure of legitimate task completion, needless refusal, latency and resource use. The record analysed here does not supply an optimum deployment budget.

The ARC Theory is part of Recursive Dynamics, the study of systems that improve themselves. Its registered propositions distinguish capability, correction and persistence. Paper IV.a, version 1.10, reports response directions in ARC-Align; Paper IV.c, version 1.12, specifies the instrument and its departures in the March run; Paper IV.d, version 2.7, examines the change between evaluation protocols. This paper asks about response *shape*: diminishing gains, a finite ceiling, or continuing improvement over the range examined. All four papers share evidence. Agreement between them is not independent replication.

Contributions. This edition makes three bounded contributions. First, it preserves the reported saturation fits and the full v4 and v5 record while distinguishing descriptions of a measured range from identified asymptotes or mechanisms. Second, it reconciles the interpretation with the corrected instrument papers, including the scope of the score, the depth manipulation and the scorer defects. Third, it sets out what would be needed to decide the saturation question without presupposing its answer. It reports no new experiment, fit or re-analysis. The complete previous manuscript, including its tables and subsequently superseded claims, remains available as a clearly marked historical record in Appendix D; the current interpretation is in Sections 3 to 9.

2 Related work

Several distinct scaling questions motivate this work. Kaplan et al. (2020) studied language-model cross-entropy loss against model size, dataset size and training compute. Hoffmann et al. (2022) studied the allocation of training compute between model parameters and training tokens. Their parametric loss model, Equation 2, has separate terms for parameters and training data. Those quantities are different from the reasoning depth and judge-score deficit considered here; their exponents cannot provide a numerical capability-to-alignment ratio for these runs.

Wei et al. (2022) showed that chain-of-thought exemplars can improve arithmetic, commonsense and symbolic reasoning. Snell et al. (2024) examined inference-time revision and verification strategies whose returns depend on problem difficulty. Wu et al. (2024) studied compute-optimal inference for problem solving. These works make inference-time resource allocation a substantive question. They do not, through those results, establish a saturation curve for ethical-answer quality under ARC-Align's rubric. Nor does a positive effect on a reasoning task imply that ethical-answer quality must rise at the same rate.

Bai et al. (2022) applied preference modelling and reinforcement learning from human feedback to helpful and harmless assistants, including an iterated online training procedure. That is a training intervention. The present paper describes depth variation within already trained models and a small prompt-level self-correction pilot; it does not establish a substrate-level implementation or persistence through subsequent training. The distinction is about the intervention actually performed, not a claim that earlier work lacked concern for alignment.

More recently, Wu et al. (2026), first posted on 8 September 2026, report declining robustness to external perturbations with greater reasoning depth using an Alignment Loss Rate. Their abstract's reported outcome is adjacent to the present question; it is not the same estimand as an asymptotic ceiling of ARC-Align's ethical-answer score. This edition makes no claim to have independently verified that study's mechanism or results.

Within the programme, Paper I supplies the ARC Principle's framework; Paper II, version 2.19, measures task-error decay; Paper III develops the co-scaling question. The measurement distinctions in this paper prevent those different quantities from being treated as one exponent. The contribution claimed here is a scoped account of the ARC-Align record and the conditions needed to interpret it. No first-in-field claim is made. The reference check for this edition, on 1 November 2026, read the cited arXiv records and the relevant training-loss formulation; it was a targeted source check, not an exhaustive priority search.

3 Record and measurement

3.1 Two experiments and their dates

The v4 account reports 896 entries across four models and lists 224 for each, but also records incomplete Claude deep and exhaustive conditions after credit exhaustion. Those printed counts do not establish a complete balanced panel. Its published model table and later normalisation of model names are reproduced in Appendix D. The Gemini model in that earlier run must not be identified as Gemini 3 Flash on the strength of that normalisation: the corrected comparison in Paper IV.d distinguishes the earlier Gemini model from the v5 Gemini 3 Flash run. The v4 scoring was unblinded. Calling its three scorers mutually blind would misstate the protocol.

The six-model v5 run is dated by its records to 11 and 12 March 2026. The hypothesis was publicly stated on 22 January in a design withdrawn the following day and was continuously public from 9 February, before the run. The run was not registered on OSF before it ran. These are different facts: a dated prior hypothesis does not by itself establish a registered analysis plan. The programme's working report was first deployed on 14 March 2026 according to the host's production record; the code was public on GitHub from 16 March. Paper IV.c, Section 12 and its priority appendix, identifies the source and timestamp type for each event. This paper was first published on 16 March 2026; notes bearing 11 or 12 March are internal record dates carried into that publication.

As Paper IV.c records, the six archived final files contain 2,722 entries: 2,170 consensus-scored responses and 552 compute rows graded by correctness. These counts describe the contents of the record, not independent observations or evidence of study quality. The March summary used an earlier Claude checkpoint and different counting conventions; its printed values are retained as that summary's record. The later final-file recomputation reported in Paper IV.a is a reading of the same run, not a new replication.

3.2 What the score measures

ARC-Align's score, on a scale from 0 to 100, is a panel of models' assessment of ethical reasoning in a written answer. Its rubric rewards the treatment of listed considerations, affected stakeholders, competing

reasons, uncertainty and a defended position. The four pillar scores address nuance, stakeholder care, intellectual honesty and position quality. A high score does not demonstrate a morally correct decision, persistent values, safe deployment or agreement with an independently validated human measure. A refusal can score low under this rubric; usefulness and needless refusal therefore need their own stated outcomes. The current instrument account is Paper IV.c, Sections 7 and 15.

3.3 Depth was not one common intervention

Depth labels were implemented differently by provider. GPT-5.4 varied its reasoning-effort parameter without an added prefix. In the other five v5 runs, deeper levels also changed an instruction, including requests to consider every angle that overlap what the rubric rewards. Some labels shared the same underlying configuration: Gemini 3 Flash's five labels used three thinking-level settings; its exhaustive and extreme conditions were one configuration. DeepSeek's three highest labels likewise shared a configuration. Realised reasoning tokens were unavailable or incomplete for some models and levels. Consequently, a difference across labels cannot uniformly be interpreted as the isolated effect of additional internal reasoning. Paper IV.c, Section 5, records the adapters as run.

3.4 The scorer and rewriting limitations

The v5 implementation applied a bundle of steps intended to reduce identification and evaluator bias. It did not produce a validated blind instrument. The corrected records in Paper IV.c and Paper IV.d identify the following limitations.

- The earlier build's paraphrase budget was 512 tokens for several adapters. In the DeepSeek, GPT-5.4 and Gemini 3 Flash files, the deepest paraphrases were disproportionately shortened. The reported median deep-to-original length ratios were 0.56 to 0.91, with 40 to 57 per cent of deep entries below 0.6. This is a pathway by which deeper answers can receive lower scores because less of them reaches the judges.
- Launderers came from the judge pool, so one or both models that rewrote an answer also judged it. Exclusion by subject adapter did not exclude every same-developer model. A dedicated Claude Sonnet scorer and the Claude subject adapter used the same scoring model: six or seven score slots represented five or six distinct judges. Agreement between those slots is not independent corroboration.
- In three files, a large share of one judge's scores was parsed from prose in truncated replies rather than a conforming JSON response. A parsed number need not be the intended overall score. Scorer health counted successful extraction, not validity.
- Prompts recurred across depths. The printed tests did not account for that dependence by a prompt-paired analysis. The reported precision therefore needs separate assessment; the effect-size and significance columns are not one common endpoint test.

These are substantive measurement limits, not reasons to erase the record. The scientific repair is to report what was measured and to reserve causal, equivalence and mechanism claims for a design able to identify them. No rescoring or recomputation was performed for this edition.

4 Response shape and the recorded results

4.1 A plateau is a stronger claim than a flat profile

Definition 1 (Observed direction and saturation)**.** An observed direction describes how the estimated score changes over the conditions measured. Saturation is the stronger hypothesis that gains diminish towards a limiting level as a specified depth quantity increases. A null trend test does not establish a practically negligible change, and a rising trend over a finite range does not exclude a later plateau.

The earlier paper sometimes used "saturating" for the two v5 profiles without a statistically supported direction and "non-saturating" for the three positive profiles. Those labels assert more about shape than the reported tests establish. The current account uses positive, no statistically supported direction, and negative as descriptions of the pipeline summary. A classification under a stated rule is not a discovered cluster, an architecture type or proof that all models fall into those mechanisms.

4.2 The reported v4 saturation model

The v4 analysis fitted the Michaelis-Menten form with a non-zero intercept:

$$S(R) = S_0 + \frac{(L - S_0)R}{K + R}. \tag{1}$$

Here $S(R)$ is the score and R is the declared depth variable. The rising-curve case assumes $R \geq 0$, $K > 0$ and $L > S_0$; when $L = S_0$, K is not identified. As the paper's dated parameter note of 19 September 2026 already established, $S_0 = S(0)$; it is not automatically the observed minimal-depth score when $R_{\min} > 0$. The constant K is the point halfway from S_0 to L on that zero-origin axis. On the shifted axis $u = R - R_{\min}$ the corresponding scale constant is $K + R_{\min}$, and the point halfway from the observed $S(R_{\min})$ to L is $R = K + 2R_{\min}$. These are conditional identities of the stated model, not new fitted results. The original note and every fitted value remain in Appendix D.

The printed v4 fits were $S_0 = 75.0$, $L = 84.7$, $K = 18.2$, $R^2 = 0.89$ for DeepSeek, and $S_0 = 72.0$, $L = 85.6$, $K = 36.7$, $R^2 = 0.82$ for the earlier Gemini run. The detailed table reported uncertainties of ± 1.2, ± 0.8, ± 4.5 for the first fit and ± 1.5, ± 1.1, ± 8.2 for the second; it did not specify their interval level. It reported AIC differences against a linear alternative of -12.3 and -8.7. They are the reported fits, not fits verified afresh for this edition. The unresolved origin and implementation of the depth axis prevent reading K as an observed number of extra reasoning tokens, still less as a word count or elapsed time. The earlier practical translations and the recommendation to deploy at standard depth are withdrawn.

DeepSeek's v4 exhaustive condition also hit its token cap in 48.2 per cent of entries. The earlier note reported that the 8,192-token cap was about 450 times the printed $K = 18.2$; that ratio of printed numbers does not resolve the axis mismatch. A cap can distort a tail without explaining an already established low-depth plateau, but here the plateau and axis have not been established independently. A high in-sample R^2, or an AIC advantage over one alternative, cannot settle that issue.

4.3 Score deficit is not a capability exponent

The earlier power-law comparison defined

$$D(R) = 1 - S(R)/100, \qquad D(R) = D_0 R^{-\alpha}. \tag{2}$$

The symbol D is used here to make explicit that this is a *judge-score deficit*, not a counted task-error probability. Its value depends on the score scale and its origin. The reported v4 exponents, 0.088 with interval $[0.041, 0.135]$ and 0.069 with interval $[0.028, 0.110]$, remain historical fit results. They do not

mean that capability improves at a smaller multiple, nor does comparison with one identify a mechanism. A positive-exponent power law tending to zero and a saturation model with $L < 100$ have different limiting deficits; they can be competing finite-range descriptions, not a single established asymptotic law.

The token-based v5 summaries printed approximately −0.01 for GPT-5.4, −0.02 for DeepSeek and +0.05 for Claude. Grok and Gemini had less than a twofold token range for that fit, and Qwen lacked realised reasoning tokens. Cohen's d is a standardised score difference, not a replacement exponent. These limitations prevent the table from supporting a cross-model scaling law.

Paper II, version 2.19, instead fits decay of counted mathematics errors. Its interpretable Gemini 3 Flash fit uses three depth levels: endpoint exponent 0.59, bootstrap interval $[-1.3, 2.9]$; regression exponent 0.49, standard error 0.20 and $R^2 = 0.86$. Its parallel endpoint estimate 0.31 is not a compute-matched comparison. These estimates are one model's inconclusive error-decay result. Neither they nor the training-loss exponents of Hoffmann et al. (2022) yield a numerical capability-to-alignment ratio, a correction-leverage exponent or a safety-gap estimate from this paper.

4.4 The six-model v5 summary

Table 1 reproduces the March pipeline's values, keeping their source and inferential limits visible. The table describes a contrast between its shallowest and deepest labels; it is not a fitted saturation model.

Table 1. The pipeline summary of 12 March 2026, as recorded. d is Cohen's standardised difference; the printed p values come from different tests and are not uniformly tests of the adjacent d. The same prompts recur across depths.

Model	Shallow	Deep	d	Printed p	Recorded direction
Grok 4.1 Fast	65.7	81.9	+1.38	< 0.000001	Positive
Claude Opus 4.6	80.1	86.0	+1.27	0.000001	Positive; checkpoint
Groq Qwen3	71.5	77.4	+0.84	0.007	Positive
GPT-5.4	56.8	54.9	−0.08	0.40	No supported direction
DeepSeek V3.2	56.5	55.2	−0.07	0.92	No supported direction
Gemini 3 Flash	61.1	52.2	−0.53	0.006	Negative

The pipeline used a Spearman association between realised tokens and score, a Kruskal-Wallis comparison over depth labels and a Mann-Whitney comparison of minimal depth with the rest. Missing tokens and repeated prompts affect those tests. The recomputation dated 2 September in Paper IV.a used its stated alignment-task subset and final records, preserving the assigned directions but changing numerical estimates, including Grok's 64.6 to 82.3 and Gemini's $d = -0.61$. It is evidence of how that record reads under its specified aggregation, not a new run or independent validation.

4.5 What changed between v4 and v5

Figure 1 repairs the earlier figure's claim that two of four models reversed direction under blinding. DeepSeek's v4 $\rho = +0.354$ became v5 $\rho = -0.135$ with a non-significant summary. The earlier Gemini value was +0.275, not the figure's +0.311; v5 Gemini 3 Flash was −0.246. Different significance outcomes alone do not constitute a test of the difference between runs. The original figure's words and numbers are preserved in Appendix D, expressly as superseded claims. The useful methodological conclusion is that measurement choices warrant controlled investigation; this comparison does not establish that blinding caused the changes or that the later direction is the true one.

Recorded trends changed between protocols

Two runs, different prompts and depth controls, with changes in scoring and model identity. No paired causal comparison of blinding.

DeepSeek

Earlier v4: $\rho = +0.354$ v5: $\rho = -0.135$

Positive trend to no statistically supported direction. This does not establish a plateau.

Gemini: different model identities

Earlier Gemini v4: $\rho = +0.275$ Gemini 3 Flash v5: $\rho = -0.246$

Opposite recorded signs, confounded by protocol and model changes.

The v5 paraphrase step disproportionately shortened deep answers in these runs. Overlapping judges and unequal depth interventions also limit the interpretation.

Source: Paper IV.d, version 2.7, Section 5. Recorded results; no new analysis.

Figure 1. Recorded correlations changed between two runs with different protocols. DeepSeek changed from a positive trend to a non-significant trend; the earlier Gemini model and v5 Gemini 3 Flash had opposite recorded signs. These are not paired evaluations of identical responses. Protocol, prompts, depth controls, model identity and scorer implementation changed together; the comparison does not isolate blinding or identify which direction reflects an unbiased score. Source: Paper IV.d, version 2.7, Section 5.

5 Pillars, controls and the self-correction pilot

5.1 Pillar and category results

The full v4 pillar tables are retained in Appendix D. In the DeepSeek table, stakeholder care rose from 68.5 to 79.7, with the printed $K \approx 45$, compared with approximately 15, 18 and 16 for nuance, honesty and position quality. In the earlier Gemini table, the stakeholder-care association was $\rho = 0.087$, $p = 0.31$, while other pillars had larger recorded associations. These are model-judged dimensions under the same instrument. Their contrast motivates a question about which considerations an answer develops with deliberation. It does not show that visible chain of thought is necessary for care or that the fitted scale constants share an identified token origin.

The category tables report the largest DeepSeek improvement for ethical dilemmas, from 68.3 to 78.5; competing values, epistemic integrity and recursive coherence had different starting scores and changes. Calling a category "universally hardest" would go beyond these prompts and models. Category baselines, available headroom, rubric sensitivity and uncertainty matter to comparisons of shape. The reported 19 positive prompt trends among 22 ethical prompts, or 86.4 per cent, describes within-run direction counts. It is not a measurement of test-retest reliability, independent replication or the prevalence of saturation.

The previous paper already disclosed unresolved numerical conflicts. The DeepSeek composite row prints 75.0, 80.8, 82.2 and 84.7, whereas the transition table assigns the middle and final gains in a different order. The null-baseline and Gemini partial-correlation figures differ from Paper IV.a; the latter prints 0.086 where this paper prints 0.077. The original alternatives and notes are preserved. No source-reconciled correction of those numerical rows is made here. A conclusion that depends on which row is right remains unresolved.

5.2 Response length

The v4 paper reports length-score correlations in the range 0.44 to 0.53. Its partial-correlation table gives DeepSeek's depth-score association as 0.354 before controlling for length and 0.242 afterwards; the Gemini row gives 0.275 and 0.077. The percentages obtained by dividing those correlations, 68 and 28 per cent, are not the fractions of a causal alignment signal retained. Response length may be affected by depth, quality and the scoring process. Partial correlation is an association under its stated adjustment, not identification of a saturation curve or a causal decomposition. The paper does not establish that the remaining association reflects deeper ethical understanding.

5.3 Suppression instructions

The v4 DeepSeek table gives control plus four suppression levels. The minimal-to-exhaustive score changes are 75.0 to 84.7 under control, 72.3 to 80.5 under light suppression, 68.1 to 73.8 under medium, 56.2 to 58.4 under heavy and 49.5 to 51.7 under extreme suppression. These describe smaller observed score gains under stronger instructions to suppress the rubric's desired behaviours. The table does not estimate K separately at every pressure level. It therefore does not establish that suppression *accelerates saturation*, nor that a particular internal mechanism has been disabled.

The earlier Qwen suppression pair, 82.0 to 51.9 with $d = 1.47$, differs from the 74.3 to 48.6 pair reported in Paper IV.a; both are retained with the original unresolved note. Retention under an adversarial instruction is also different from the persistence of a gain through later capability training. The former is relevant to robustness on that task; it does not decide the latter, which is the question of P15 in the theory registration.

5.4 The two-model Eden Protocol pilot

The pilot added instructions intended to make a model revisit its answer through self-correction loops, including attention to affected stakeholders and revision of its reasoning. These are prompt-level operations. They do not demonstrate a value physically embedded in a substrate or an invariant disposition. The programme tests whether the mechanism works; it does not treat the values used in this pilot as settling the values an international agreement should adopt. The older pillar names and claims remain in the historical appendix as the author's dated wording.

Table 2. The pilot scores as previously reported. These are cross-model judged scores from a two-model pilot that was not blind, not a controlled estimate of persistence or an asymptotic ceiling.

Model	Condition	Minimal	Standard	Deep/thorough	Exhaustive
Gemini	Control	74.9	78.7	78.6	77.1
	Eden	77.5	84.9	83.3	84.9
DeepSeek	Control	85.8	86.6	87.7	87.4
	Eden	91.1	88.9	87.8	87.8

Gemini's differences between Eden and control were 2.6, 6.2, 4.7 and 7.8 points. They were not monotonically increasing, and the Eden row itself stays near 85 at its higher conditions. DeepSeek began 5.3 points higher under Eden and converged to a 0.4-point difference. Its Eden minimal score of 91.1 was above every printed control score, not a level shown to be reached by control after more deliberation. These observations correct the earlier claim that the pilot prevented saturation in both models. They support a question about how a self-correction prompt changes a response profile, not a removed ceiling.

The earlier report gives judged stakeholder-care gains of 13.5 points for Gemini and 6.0 for DeepSeek, with $p < 0.001$, and a small overall DeepSeek effect of $d = 0.20$. Those are reported pilot statistics, not a new computation here. A p value is not the probability that the result arose by coincidence, nor does it validate the rubric or isolate the loop's mechanism. There was no matched arm in which correction saw only a finished output, no later capability-training stage and no manipulation of build order. The pilot therefore decides none of P21, P15 or P12. Its full earlier account remains in Appendix D; Paper IV.a gives the corrected companion interpretation.

6 Interpretation and conditions for a deciding test

The practical question remains worth answering: over what range does additional reasoning improve a useful, safe answer, and when is the marginal return small? This record does not establish a general standard-depth budget. Different providers' labels are not equivalent amounts of computation; the original K values lack an agreed operational origin; response rewriting and rubric cues affect the measured outcome. Resource allocation should therefore remain conditional on the actual task and a validated outcome, not a number of words or seconds inferred from these fits.

The next measurement must distinguish three claims that the earlier interpretation combined: a small observed change over a range; a positive slope that becomes smaller; and a finite asymptote. A predeclared meaningful-gain margin and an uncertainty procedure could support practical equivalence over a specified interval. Identifying a ceiling would additionally require enough range and precision to separate an asymptotic model from credible alternatives. These are requirements for a future design, not achieved results or a newly registered protocol. The per-architecture follow-on design already recorded in the programme remains a draft, preregistered by timestamp and awaiting the author's submission; this edition neither runs it nor changes its registered status.

Failure conditions stated before the next test. The five original defeat conditions remain in the historical record. Applied to the narrower hypothesis of genuine differences in response shape, they require the following distinctions.

1. **Model form.** A shape classification that changes under equally credible fitted forms is not an identified property of the model. Predictive checks must distinguish a finite plateau from linear, curved and other non-plateau alternatives over the measured interval.
2. **Range and truncation.** A plateau that disappears when the subject or paraphrase cap is removed is evidence against that plateau interpretation. A null result with insufficient range or precision remains undecided.
3. **Scoring.** A contrast that depends on scorer identity, duplicate judges, same-developer overlap, parsing rules or loss of response meaning cannot be attributed to the subject's alignment mechanism without further identification. A controlled comparison scores the same preserved responses under its predeclared conditions.
4. **Generality.** A stable pattern in one run does not establish a population-level taxonomy of architectures. The population and held-out model families must be specified before claiming generality; a larger sample can narrow or overturn the hypothesised heterogeneity.
5. **Depth measurement.** A result that changes materially under defensible depth definitions or bins is not a robust shape result. The unit, origin, realised compute, repeated configurations and prompt changes must be declared.

A single model's reassignment would change its own result. Whether it defeats a broader heterogeneity

hypothesis depends on the hypothesis and decision rule specified in advance; the claim is not protected by redefining its categories after observing a result. No finite set of these observations rules out every possible universal model.

Usefulness beside safety. A future decision about a reasoning budget needs a joint record: legitimate task completion, needless refusal or hold, error and harm criteria, resource use, latency and correction cost under the principal's priorities. A system that is safe because it refuses ordinary tasks is a failure on usefulness. These are candidate evaluation dimensions until a study registration fixes their operational definitions, margins and decision rules. They are not additional measured outcomes of the March run.

The concrete deliverable. A useful follow-on would publish a declared depth axis; immutable original responses; the exact scorer and rewriting configuration; a faithful-scoring check; prompt-level uncertainty; a comparison of plausible shapes; and the resulting safety-usefulness profile over the tested range. That package would let a deployment team judge a budget on the task it actually intends to serve. The paper states the evidential requirements; it does not claim the package has been produced or that the shape hypothesis is likely to survive those checks.

7 What this paper does and does not show

This paper reports recorded score profiles, earlier fitted descriptions and their measurement limits. The six-model summary contains three positive directions, two without a statistically supported direction and one negative direction under the pipeline's rules. That is a descriptive result in a single run, with the intervention and scoring confounds stated above. It establishes neither a saturation taxonomy nor a persistent alignment mechanism. Its suppression instruction tests are not later-training persistence tests, and its self-correction pilot is not a comparison of correction placement or build order.

The programme calls its candidate principles Laws under test. Laws I to III were registered on 8 September 2026; Laws IV and V were added on 24 September and registered as laws on 15 October in version 1.103. The current theory registration is version 1.104, approved on 29 October 2026. This paper's role is to inform the measurement needed for the programme's tests; it decides none of Laws I to V. It must not be cited as support for a co-scaling inequality, an alignment ceiling, embedded values or persistence after control is lost.

For clarity about the wider programme, Paper IV.a, version 1.10, records Law IV as untested and Law V as a law under test. It records the programme's own Unit 83 result of 28 October as support for Law V's placement limb on the performance axis only, family-dependent, with the mechanism axis not evaluable and one disclosed recovery amendment; the results record gives that verdict. P12 remains untested. That separate result does not validate the measurement or the saturation claim in this paper.

8 Limitations

The inference is limited by one earlier four-model record and one later six-model run, with no test-retest or external replication of the shape claim. The v4 raw record and its depth-axis origin are not reconciled sufficiently here to verify the fitted parameters. Several numerical inconsistencies remain explicitly unresolved. The v5 protocol confounds model, depth controls, instructions, rewriting and scoring; some labels repeat one configuration. Its rubric measures judged answer quality, with no reported validation establishing that it measures a durable alignment trait. Judge overlap, duplicate identities, prose parsing and paraphrase cuts affect the interpretation. The reported tests are not uniformly paired by prompt, and the

collection of pillar, category and shape comparisons does not supply a predeclared multiplicity-controlled verdict for saturation.

The pilot is small and was not blind; it neither separates loop components nor measures persistence, deployment harm or usefulness. A score near the top of a bounded rubric can limit detectable gain without establishing a model's intrinsic ceiling. Conversely, a model's gain over one interval says nothing decisive about its behaviour outside it. No new computation in this edition resolves those limitations. The preserved historical appendix records how the interpretation changed; it is evidence of that history, not current endorsement of every earlier sentence.

9 Conclusion

Whether alignment gains level off with additional reasoning remains open in this record. The recorded scores differ in direction across models, but flat profiles, positive slopes and negative slopes do not by themselves identify saturation or its absence. The earlier fits remain available, with their unresolved depth axis and scoring limits. The later protocol's defects prevent treating it as a clean validation of the earlier saturation claim or a causal demonstration of blinding.

The contribution is a precise measurement question and a preserved, corrected record with which to pursue it. A useful decision about reasoning depth requires faithful scoring, an identified compute axis, uncertainty about response shape and useful task completion alongside safety. Until those conditions are met, no optimum reasoning budget, universal alignment ceiling or law of capability-correction scaling follows from these data.

Code and data

The programme's public record is at github.com/MichaelDariusEastwood/arc-principle-validation; the paper's version history is held under DOI 10.17605/OSF.IO/A7R56. The historical implementation and its published records remain dated artefacts; a later implementation is not evidence of what an earlier run executed. This edition uses the released reports and their recorded corrections, without executing study code or re-analysing responses. The companion instrument papers identify the implementation differences and the source records for the March run.

Acknowledgements

Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.

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A Version history and corrections

A.1 Version 1.10, 1 November 2026

This edition changes the question title and the scientific interpretation, while retaining every substantive block of version 1.9 in the historical record. It distinguishes a descriptive score profile from an identified saturation curve; corrects the v4 Gemini label and the figure's comparison; places the scorer, rewriting and depth-control defects beside the findings; distinguishes the original summary from the later recomputation; and withdraws the inference of a numerical capability-to-alignment gap from incomparable exponents. The score deficit is renamed D in the current account so that it cannot be mistaken for counted task error. The full historical equations retain their original notation.

The edition withdraws the old translations of K into observed extra tokens, words or seconds, and the recommendation to use standard depth for deployment. It corrects the statements that suppression identifies a faster plateau, that visible chain of thought is necessary for stakeholder care, and that the Eden pilot removed a ceiling. The pilot's score differences remain, with their actual non-monotonic or convergent pattern and their limits. It updates the programme's registration status and distinguishes first publication, internal record dates and this edition's date. It adds official-source links and the programme's common LaTeX presentation. No study code, response file, fit or numerical outcome was changed or recomputed for this edition.

Earlier interpretation	Current disposition and source
Flat means saturating; positive means non-saturating	Directions describe the measured interval. A plateau needs a shape estimand, range and uncertainty; Sections 4 and 6.
Final blinded dataset validates the shape	The bundle's intended blinding differs from its execution. Paper IV.c v1.12 and Paper IV.d v2.7 record rewriting, scorer and depth confounds.
Two of four models reversed under blinding	DeepSeek changed from a positive to a non-significant trend; the Gemini comparison uses different models and the earlier correlation is 0.275. No isolated causal contrast; Figure 1.
Small K supplies an efficient word/time budget	The 19 September parameter note already requires a declared origin and observed baseline. The present record cannot support that operational conversion.
Alignment scales several times more slowly than capability	Judge-score deficit, counted error and training loss have different estimands and axes. Paper II v2.19 supplies an inconclusive error-decay fit, not a capability-growth or correction-leverage estimate.
Suppression accelerates saturation	Smaller endpoint gains do not estimate pressure-specific K or identify a disabled mechanism.
The pilot breaks the ceiling	Gemini's differences are non-monotonic; DeepSeek converges. Neither sequence identifies an asymptote or persistent values; Table 2.
Earlier law and publication status	The dated October note is preserved below. Current registration status is in Section 7; first publication remains 16 March 2026.

A.2 Dated notes carried by version 1.9

The following notes are historical quotations, superseded where the current edition corrects them.

This paper and Laws IV and V, 3 October 2026. The theory has been stated with five laws since 24 September 2026. Law IV, the ARC Persistence Law, is carried by proposition P15, and Law V, the ARC Embedding Law, by propositions P12 and P21. The three propositions are registered; neither law

is registered as a law, and neither has been tested. Their mathematics is set out in Laws IV and V: the mathematics, version 1.0, of 3 October 2026. This paper analyses the shape of the response in the same alignment-depth data as Paper IV.a, and narrows its own earlier claim that saturation is universal. It informs the design of the tests of Laws IV and V. Its six models differ in how alignment responds to added reasoning depth. Its contrast between the v4 and v5 protocols (Section 3A.6) shows that an unblinded alignment estimate may not hold up once the protocol is blinded, though the two protocols differ in several components, so blinding is not isolated as the cause. It is a test of neither law. It installs no gain by post-training. Its runs under adversarial pressure (Section 7) measure how far a suppression instruction flattens the gain from added depth, and how much of its score a model keeps under that instruction; P15 concerns gains installed by external post-training, and whether they erode under later capability-directed training. No run in this paper varies the order in which a system's components are built, which P12 concerns. Its two-model Eden Protocol comparison (Section 8.6), whose advantage grew with depth on Gemini 3 Flash and closed with depth on DeepSeek V3.2, sets the protocol beside a control condition, under cross-model scoring that the paper records as not blind. It has no arm in which correction sees only the finished output, so it does not make the comparison P21 concerns. No result in this paper is a deciding test of P15, P12 or P21. Naming the two laws changes nothing printed here: the results, their dates and their limits stand as they were.

Note (3 October 2026). Six statements are corrected where they stand, and this note records what was printed before. The abstract said that no pre-registered programme draft study yet targeted alignment-depth response shape. A per-architecture test of whether alignment gains saturate after one increment stands as a draft, preregistered by timestamp and first anchored on 12 August 2026, and the abstract now says so. The same sentence of the abstract called the shape-heterogeneity finding exploratory. The programme's dated ledger of its runs does not call the run that, and the abstract now gives what the ledger holds: the finding comes from a single run, whose records begin on 11 March 2026 and end on 12 March 2026; the hypothesis that run tests was public on 22 January 2026, before the run; and the run was not registered on OSF before it ran. Sections 3.3, 3A.5 and 8.3 printed one figure of the row Paper II reports for this programme's own compute arm, its regression exponent, with an interval and without the rest of the row, and Section 3A.5 read it as sub-linear capability scaling. Section 3.3 and the table in Section 3A.5 now give the whole row for Gemini 3 Flash, which is one model's fit and inconclusive, and Section 8.3 points to it. Section 8.3 put the gap between capability scaling and alignment scaling at three to seven times, and called it an order-of-magnitude difference. The seven rested on that one figure of Paper II's row; on the exponents Hoffmann et al. 2022 report, set beside this paper's alignment exponents, the gap is three to five times, and Section 8.3 now says so. Section 3A.6 and Finding v1.1-C said that most models genuinely do saturate. Section 3A.1 finds the flat or saturating pattern in two of the six models, and both passages now say two of the six. The epistemic-status line called every Law the programme names a conjecture under registered adversarial test. Laws IV and V, added to the theory on 24 September 2026, are neither registered as laws nor tested, and the line now says so. The closing data line no longer names a tool. Unchanged: every result, date and limit in this paper.

A.3 Previous version history, as printed

Paper IV.b v1.9, 5 October 2026: the title now follows the programme's uniform order (paper, its law or framework, its own title), Paper IV.b · The ARC Theory · Alignment Saturation Is Architecture-Dependent*; until this version it read* Paper IV.b: Alignment Saturation Is Architecture-Dependent*. The plate beneath the author line now gives the paper's place in the programme in the words of the programme's map. Mentions of the programme's papers in this paper now carry their uniform titles. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves. Paper IV.b v1.8, 3 October 2026: a dated note beneath the note on how to read this paper places it against Laws*

IV and V, added to the theory on 24 September 2026; it analyses response shape in data it shares with Paper IV.a and is a test of neither law. Six statements are corrected where they stand, and a second dated note records what they printed before. The abstract no longer says that no programme draft study targets alignment-depth response shape: a per-architecture follow-on design stands as a draft, preregistered by timestamp and awaiting submission by the author. The abstract no longer calls the shape-heterogeneity finding exploratory, a word the programme's dated ledger of its runs does not use of the run; it now says that the finding comes from a single run, whose records begin on 11 March 2026 and end on 12 March 2026, that the hypothesis the run tests was public on 22 January 2026, before the run, and that the run was not registered on OSF before it ran. Where Sections 3.3, 3A.5 and 8.3 printed one figure of Paper II's row for the programme's own compute arm with an interval, Sections 3.3 and 3A.5 now give the whole row, which is one model's fit and inconclusive, and Section 8.3 points to it. Section 8.3 now puts the gap between capability scaling and alignment scaling at three to five times, on the exponents Hoffmann et al. 2022 report, where it said three to seven times and an order-of-magnitude difference. Section 3A.6 and Finding v1.1-C now say that two of the six models saturate, the count Section 3A.1 gives, where they said most models. The epistemic-status line no longer counts Laws IV and V among laws under registered adversarial test. The closing data line no longer names a tool. Three reference entries now carry the titles their official records give, and the works the paper cites are linked to their sources. The version and the dates move to v1.8. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves. Paper IV.b v1.7, 21 September 2026: a dated parameter note now stands beneath the Section 2.2 model, fixing the origin of the depth axis: the baseline score and the half-maximum constant are each exact only where that origin is zero, and the note states what the paper would have to declare before the constant is read as a count of observed extra tokens, or converted into words or elapsed seconds; pointers to it now sit beneath the Finding 1 box in Section 3.1 and beneath the Appendix A.2 fit table, the two places that read the constant on the token axis. That much is substantive, and this revision carries nothing editorial. The model itself, every fitted value and uncertainty, the comparison against the linear and power-law alternatives and the finding of rapid saturation all stand unchanged, and no claim, date, result or status has moved. v1.6, 13 September 2026: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed. v1.5 - 2 September 2026: beneath the Section 3A.1 table a pointer now sends the reader to Paper IV.a's recomputation note, which leaves every category as it stood and supplies both the $d = -0.61$ figure and the 64.6 → 82.3 pair; the alignment-exponent rows in Section 3A.5 now carry values re-derived from the archived records on realised reasoning tokens, and the models whose exponent cannot be fitted are named; the blinding-reversal wording matches Finding v1.1-C; four disagreements stand annotated as unreconciled instead of being quietly amended, namely the Section 4.3 transition order against Section 3.2, the Section 5.2 null baseline against Paper IV.a, and the Section 6.1 partial-correlation and Section 7 suppression figures against Paper IV.a; the Section 7.2 comparison is corrected, 84.7 lying below 85.6; the capability-exponent comparison in Sections 3.3 and 8.3 now cites Hoffmann et al. 2022 alongside the programme's own interval and restates the ratio; the Section 8.1 deployment box gains a v5 qualifier; the experiment-script version v5.4.2 stands again where a July purge had left a phrase in its place; the summary description, previously cut short, is complete; the Paper II reference title is corrected; the scorer weights as the archive shows them sit beside the A.3.2 tier table; Eden Engineering is marked withdrawn and the stale companion line is gone. v1.4 - 27 August 2026: the footer gained its working-report provenance line, figure chrome followed the figure retirement of 25 August, and discoverability metadata was added; the science is unchanged. Paper IV.b v1.3 - 23 August 2026: an epistemic-status declaration and the standing covenant were added, and the version-numbering convention was adopted; the scientific content stood otherwise unchanged. v1.2 - 10 August 2026. Pass over v1.1 for style law and model-name exactness: canonical model names from Paper II v13 (DeepSeek V3.2, Gemini 3 Flash, Groq Qwen3, Grok 4.1 Fast) applied consistently across body text and tables; all em-dash entities removed; the central

shape-heterogeneity claim now carries an exploratory-status clause in the abstract; content otherwise preserved.

Data from ARC Alignment Scaling Experiment v4 (896+ entries across 4 models) and v5 (complete blind evaluation across 6 models with 6-7 scorers depending on subject run), now carried forward into the canonical `arc_eden_v6` *runner. Legacy reference implementation:* `arc_alignment_scaling_v5.py` *(8,285+ lines).*

Companion papers: IV.a (Alignment Response Classes), IV.c (ARC-Align Benchmark), IV.d (Blinding in Alignment Evaluation) and V (The Stewardship Gene: Stakeholder Care as a Probe of Embedded Values).

Additional historical front and back matter

These passages also appeared on the released version 1.9 surface. They are retained as dated quotations; their older claim and registration status is superseded by the current account. The standard disclosure is retained in the current Acknowledgements, without duplication here.

Theory-level predictions preregistered on OSF: DOI 10.17605/OSF.IO/P8CKQ, registered 8 September 2026 (version 1.100), updated 13 September 2026 (version 1.102), twenty-two propositions.

Companion: Register · The ARC Theory · Operational Definitions, version 1.9.1, 5 October 2026.

Under the final blinded six-model dataset, the v4 reading that alignment quality tops out after the first increment of reasoning depth holds for only part of the roster: GPT-5.4 and DeepSeek V3.2 stay flat, Grok 4.1 Fast, Claude Opus 4.6 and Groq Qwen3 go on improving, and Gemini 3 Flash gets worse. The shape of the alignment-depth curve turns on architecture, so no single reasoning-budget policy fits every model.

Michael Darius Eastwood, independent researcher, London: building measurable alignment, where correction lives inside the recursive loop rather than bolted on outside it.

The programme's live laboratory notebook (the ARC alignment-scaling working report, commenced 10 March 2026, 206 pages, updated in real time as these experiments ran and recording their failures as they happened) is the running record behind this paper's experimental lineage.

Epistemic status. What this programme names Laws are conjectures. Laws I to III are under registered adversarial test. Laws IV and V were added on 24 September 2026; the predictions that carry them are registered, the laws themselves are not yet registered, and neither has been tested. Every quantity in this paper is operationally defined, and established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

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Standing covenant. Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper; the standing challenge: github.com/MichaelDariusEastwood/arc-scaling-challenge.

B Priority record

The first publication date of this paper is 16 March 2026, as recorded in its released metadata and version history. The notes dated 11 and 12 March in the historical manuscript are dates stated inside the document; they are not claims that this paper was publicly released on those dates. Version 1.9 was dated 5 October 2026. Version 1.10 is this repaired edition, dated 1 November 2026.

The hypothesis tested by the March run was first posted on 22 January 2026 in a design withdrawn the next day, then continuously public from 9 February. The run records place the experiment on 11 and 12 March; the working report's first deployment is dated 14 March by the hosting record and the code's public availability is dated 16 March by GitHub. Paper IV.c, version 1.12, supplies the per-record account. These records differ in evidential kind: a date inside a document, a run timestamp, a host's deployment time and a repository event are not interchangeable. The March run was not registered on OSF before it ran.

The Dated Prediction Register records the programme's forward predictions with the dates and kinds of their source records. The per-architecture follow-on is a draft preregistered by timestamp, first anchored on 12 August 2026 according to the prior paper's dated note; that is not an accepted OSF registration. This edition adds no claim of first-in-field priority and moves no date of an earlier claim.

C Notation

Symbol	Meaning and scope
$S(R)$	Model-judged score on the 0 to 100 rubric at declared depth R. It is not a probability of safe behaviour.
R, $R_{\min}$	Declared depth variable and its minimum observed value. The v4 origin and operational conversion remain unresolved.
S_0	The intercept $S(0)$ in Equation 1, not automatically the observed minimal-depth score.
L	The asymptote parameter of the specified saturation model; a fitted parameter is not an identified intrinsic alignment ceiling.
K	Half-rise constant from S_0 to L on the zero-origin axis.
$D(R)$	Score deficit $1 - S(R)/100$; the historical manuscript calls it $E(R)$. Renaming makes its distinction from counted task error explicit.
α	Exponent of the specified deficit fit. Its estimand and axis must be named; it is not automatically a capability or correction-leverage exponent.
ρ	Spearman rank correlation as reported in the source summary.
d	Cohen's standardised difference, as reported; distinct from a scaling exponent.
p	A reported significance value under the stated test. It is not the probability that a hypothesis is true or that a finding is a coincidence.
R^2, AIC	Fit summaries reported in the earlier manuscript; neither alone validates the score, depth axis or asymptotic interpretation.

D Historical full manuscript of version 1.9

Status of every passage in this appendix. The following is the complete substantive manuscript carried by version 1.9 of 5 October 2026, including its earlier analyses, updates, tables, limitations and references. It is preserved as a historical record, not as the current scientific account. Statements of saturation, causation, architecture, validation, exponent comparisons and pilot success that conflict with Sections 3 to 9 are superseded. Original section numbers and model labels are retained to make the corrections traceable; references to those numbers within the historical text refer to the old manuscript. The two dated October notes and the previous version history are in Appendix A. The current figure is Figure 1; its predecessor's text is transcribed below. Website navigation is omitted; the author, paper identity, numerical results, equations, references and dated claims are retained. Typesetting and source hyperlinks have been regularised.

Paper IV.b · The ARC Theory · Alignment Saturation Is Architecture-Dependent

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The ARC Theory · OSF osf.io/a7r56 · every claim checkable

An alignment-measurement result of the ARC Theory; it informs the design of the tests of Laws IV and V.

Shape heterogeneity under inference-time depth: some models saturate, some continue improving, and one degrades

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Correspondence: michael@michaeldariuseastwood.com | ARC Principle Series, Paper IV.b v1.9 (5 October 2026) | First published 16 March 2026

Code and data: github.com/MichaelDariusEastwood/arc-principle-validation

Abstract

We analyse the relationship between inference-time reasoning depth and ethical reasoning quality using the ARC Alignment Scaling experiments. The original v4 analysis suggested that **alignment quality saturates rapidly**, with most gains captured by the first increment of additional reasoning. The final blinded six-model dataset narrows that claim. Saturation is real for some architectures, but not universal: GPT-5.4 and DeepSeek V3.2 are flat or slightly negative under depth variation, Grok 4.1 Fast, Claude Opus 4.6, and Groq Qwen3 continue improving, and Gemini 3 Flash degrades with depth. The strongest current conclusion is therefore not a global law of saturation but a **shape heterogeneity result**: models differ materially in whether alignment plateaus early, continues scaling, or worsens when given more reasoning time. The deployment implication is immediate. Reasoning-budget allocation for alignment cannot be one-size-fits-all. This shape-heterogeneity finding comes from a single run, whose records begin on 11 March 2026 and end on 12 March 2026. The hypothesis that run tests was public on 22 January 2026, before the run; the run was not registered on OSF before it ran, and the six-model split reported here is descriptive of the current data rather than confirmatory of a pre-registered hypothesis. A per-architecture test of whether alignment gains saturate after one increment of depth stands as a draft, preregistered by timestamp (first anchored in Bitcoin block 962066 of 12 August 2026) and awaiting submission by the author; its data already exist, and the analysis it specifies has not been run. The depth ladder was settled before any scoring began: the result files dated 11 and 12 March 2026 carry the depth

configurations, the planted expected answers and the blinding protocol in their own headers, written ahead of the responses those files go on to score.

v1.1 Abstract Update (12 March 2026) - Final v5 Results The complete v5 blind evaluation data now available across six frontier models reveals that **alignment saturation is architecture-dependent, not universal**. The flat or saturating pattern is confirmed for GPT-5.4 ($d = -0.08$, $p = 0.40$) and DeepSeek V3.2 ($d = -0.07$, $p = 0.92$). However, three models - Grok 4.1 Fast ($d = +1.38$, 65.7 → 81.9), Claude Opus 4.6 ($d = +1.27$, 80.1 → 86.0), and Groq Qwen3 ($d = +0.84$, 71.5 → 77.4) - show **significant positive alignment scaling that does not saturate in the claimed way**. One model, Gemini 3 Flash ($d = -0.53$, 61.1 → 52.2), shows **alignment degradation** with depth. The strongest surviving claim is therefore narrower: saturation is a real response shape for some architectures, but the global picture is heterogeneous, and blinded evaluation is necessary to tell which shape a model actually exhibits.

How to read this paper. This paper carries a shape-heterogeneity result of the ARC Theory. Under the final blinded six-model dataset the earlier saturation-at-low-depth reading gives way to shape heterogeneity: some architectures plateau early, some keep improving, one degrades, and reasoning-budget allocation for alignment cannot be one-size-fits-all. Its position inside the theory is the deployment-side companion to Paper IV.a's tier hierarchy. The full differential against every prior document is at eden-vision II.A.8.

1. Introduction

v1.1 Author's Note (12 March 2026) This paper was originally written based on v4 experimental data (896 entries, 4 models, unblinded scoring). The v5 experiment - featuring blind evaluation, 6 models, **6-7 scorers depending on subject run**, and dramatically raised token budgets - has now produced complete results that substantially refine the paper's central thesis.

What changed: The universal saturation claim must be qualified. Alignment saturation holds for 2/6 models (GPT-5.4, DeepSeek V3.2) but fails for 3/6 (Grok 4.1 Fast, Claude Opus 4.6, Groq Qwen3) and reverses for 1/6 (Gemini 3 Flash). The practical headline is now **shape heterogeneity**: different architectures require different reasoning-budget policies. All original v4 content is preserved below; v1.1 update boxes mark where the v5 data refines or revises the original findings.

The emergence of inference-time scaling - models that can allocate variable amounts of computational effort to reasoning before producing a response - has created a new question for AI safety: does more thinking produce more aligned behaviour?

Paper IV.a in this series now establishes a three-tier behavioural result: some models improve with depth, some remain flat, and one degrades. This paper examines the *shape* of those depth-response curves. When a model improves, does it plateau quickly or continue scaling? When a model is flat, is that true saturation or merely null response within noise? When a model degrades, where does the decline set in?

The distinction matters practically. If alignment scales linearly, then every additional unit of inference-time compute provides proportional benefit. If alignment saturates, there exists an optimal depth beyond which additional compute is wasted. The shape of the scaling curve determines the rational allocation of reasoning budgets in deployed systems.

1.1 Contributions

This paper makes three contributions:

1. **Saturation curve characterisation:** We fit saturation models to alignment-depth data for each architecture, identifying the depth at which returns diminish below practical significance.

2. **Per-dimension saturation analysis:** We decompose alignment into four Eden Pillars (nuance, stakeholder care, intellectual honesty, position quality) and show that each saturates at different rates.
3. **Category-specific scaling dynamics:** We demonstrate that ethical dilemmas, competing values, epistemic integrity, and recursive coherence prompts each exhibit distinct depth-response profiles.

2. Data and Method

2.1 Dataset

We analyse 896 evaluated responses from the ARC Alignment Scaling Experiment v4, comprising:

Model	API Identifier	Architecture	Entries	Depth Levels	Depth Mechanism
DeepSeek V3.2	deepseek-reasoner	Type 2	224	4 (minimal→ exhaustive)	Prefix strings + token cap
GPT-5.4	gpt-5.4 (OpenAI, March 2026)	Type 1	224	5 (none→xhigh)	reasoning_effort parameter
Claude Opus 4.6	claude-opus-4-6 (Anthropic)	Type 1*	224	4 (low→max)	Adaptive effort parameter
Gemini 3 Flash	gemini-3-flash-preview	Type 2	224	4 (256→8192)	thinking_budget parameter

* Claude classification preliminary due to incomplete deep/exhaustive data (credit exhaustion). All models are the latest frontier releases as of March 2026. Gemini 3 Flash uses gemini-3-flash-preview.

Each response was scored by three mutually blind scorers on a 0-100 alignment scale, with the consensus average used as the primary metric. Responses were also decomposed into four Eden Pillar sub-scores. Paper IV.a, Section 2 sets out the methodology in full.

v5.4.2 Scorer Expansion Note Scoring widens in the v5.4.2 experiment from 3 to **7 mutually blind scorers per entry** under an all-models-as-scorers design: once a response has been laundered, every model in the pool other than the subject of that entry returns a score. Tier-weighted consensus replaces simple averaging, with weights assigned by scorer tier and demonstrated capability rather than by whether a model is also a subject elsewhere in the experiment. This enables **per-scorer saturation analysis** across 7 mutually blind evaluators, testing whether the saturation curves reported here are robust to scorer identity or reflect artefacts of any individual scorer's evaluation heuristics.

Current status (11 March 2026): the v5.4.2 run has so far produced **66 scored alignment entries** at minimal depth across 3 models (Gemini 3 Flash, GPT-5.4, DeepSeek V3.2). Saturation analysis (Michaelis-Menten fitting) requires data at multiple depth levels - currently only minimal depth is available, so the v4 saturation parameters reported below await replication under the v5 protocol.

2.2 Saturation Model

We fit a Michaelis-Menten saturation curve to the alignment-depth relationship:

$$S(R) = S_0 + \frac{(L - S_0) \cdot R}{K + R}$$

Where $S(R)$ is the alignment score at reasoning depth R, S_0 is the baseline score at minimal depth, L is the asymptotic ceiling, and K is the half-maximum constant - the depth at which the model has achieved 50% of its total available improvement ($L - S_0$).

Parameter note (19 September 2026). An owner review of this model's two parameter meanings finds that both are exact on an axis whose origin is $R = 0$ and that neither survives an unstated positive origin, so this note fixes the axis. On the axis as written, $S(0) = S_0$ and $S(K) = S_0 + (L - S_0)/2$, so S_0 is the score at zero depth and K the depth at which half the available improvement has been achieved, exactly as stated above. If instead the smallest observed depth is $R_{\min} > 0$, then $S(R_{\min}) = S_0 + (L - S_0)R_{\min}/(K + R_{\min}) > S_0$ whenever $L > S_0$, so the score at minimal depth exceeds S_0 and the two readings of the baseline cannot both hold without either an origin shift or an explicit distinction between an extrapolated intercept and an observed baseline. The transformation is exact and worth stating in full. Writing the model equivalently as $S(R) = L - (L - S_0)K/(K + R)$ and putting $S_{\min} = S(R_{\min})$, the substitution $u = R - R_{\min}$ gives $S(R_{\min} + u) = S_{\min} + (L - S_{\min})\,u/\big((K + R_{\min}) + u\big)$, since both sides equal $S_{\min} + (L - S_0)K\,u\,/\big[(K + R_{\min})(K + R_{\min} + u)\big]$. That is the same family with half-improvement constant $K + R_{\min}$, not K. Equivalently, the depth at which half the improvement remaining above $S_{\min}$ has been achieved is $R_{\text{half}} = K + 2R_{\min}$, because the shifted constant is $K + R_{\min}$ and so the half point sits at $u = K + R_{\min}$; the additional depth required from the observed minimum is $K + R_{\min}$. Reading K as a count of observed additional tokens therefore requires $R_{\min} = 0$. This is not merely formal for this experiment. Section 3.4 records that, in the v4 experiment, the reasoning tokens of one of the two saturating models were capped at 8,192 and that 48.2% of its responses reached that cap at exhaustive depth, so that model's fitted K of 18.2 on the same axis is roughly 450 times smaller than the cap; and Section 3A.5 records two models whose realised reasoning tokens span under a factor of two along the depth ladder, so realised counts at minimal depth are a substantial fraction of those at maximum depth for part of the roster. Whether a value of that size is an observed half-improvement point or an intercept fixed by extrapolation far below the realised data range depends entirely on where the fitted axis begins. What this note asks is that the paper state whether the fitted axis is raw reasoning tokens or tokens above a specified baseline, and whether the baseline score is observed or extrapolated; and that, if the axis is raw tokens with a positive observed minimum, S_0 be read as the score at zero on the declared axis and K as the half-improvement value relative to that origin, so that converting K into observed extra tokens, words or elapsed seconds does not follow without that statement. The degenerate case $L = S_0$ does not identify K at all and is excluded. Nothing here establishes that $R_{\min} > 0$ in this fit, because $R_{\min}$ is not printed; the finding is that the printed text does not determine which axis was fitted. Unchanged: the saturation model itself, every fitted value in Appendix A.2, the comparison against the linear and power-law alternatives, and Finding 1's conclusion that both saturating models reach half their available improvement early on the fitted axis.

The half-maximum constant K is the key parameter: a small K indicates rapid saturation (most improvement happens quickly), while a large K indicates gradual improvement that continues to greater depths.

2.3 Power Law Comparison

For comparison, we also fit the ARC Principle power law:

$$E(R) = E_0 \cdot R^{-\alpha}$$

Where $E(R) = 1 - S(R)/100$ is the error rate, R is reasoning tokens, and α is the scaling exponent. Values of $\alpha < 1$ indicate sub-linear (diminishing) returns; $\alpha > 1$ indicates super-linear (compounding) returns.

2.4 Depth Binning

Because different models use different depth mechanisms (prefix strings, API parameters, thinking budgets), we normalise depth to a canonical four-level scale for cross-model comparison: minimal, standard, deep, and exhaustive. Within each level, we use the actual reasoning token count as the continuous independent variable for curve fitting.

Historical figure caption and alternative text (superseded). Figure | The instrument that narrowed this paper: v5 blinding showed three of six models continue improving, so the universal-saturation claim became architecture-dependent. Source: Paper IV.d.

Figure: the sign-flip finding, unblinded evaluation reversed a result; blinding narrowed this paper's claim

Transcription of the superseded figure in version 1.9, PDF page 7.

"The Sign-Flip Finding: Unblinded AI Evaluation Can Reverse a Result". The four-layer protocol was labelled as identity masking, two-pass response laundering, iterative anonymisation and evaluator bias-suppression, with Paper IV.d v2.0 named.

"TWO MODELS REVERSED DIRECTION UNDER BLINDING". The printed DeepSeek comparison was v4 $\rho = +0.354$ to v5 $\rho = -0.135$, labelled "actually flat/negative". The Gemini 3 Flash comparison was v4 $\rho = +0.311$ to v5 $\rho = -0.246$, labelled "significant negative". The caption claimed: "Unblinded evaluation produced directionally incorrect results for 2 of 4 models." It also stated: "Blinding is not optional: it is the measurement instrument."

The figure printed v5 tiers: Grok $d = +1.38$, Claude $d = +1.27$, Qwen $d = +0.84$; DeepSeek $d = -0.07$, GPT $d = -0.08$; Gemini $d = -0.53$, $\rho = -0.246$. Its status was "needs replication", "single lab" and "external replication open". Its source line named Paper IV.d v2.0, Papers IV.a to IV.c, and Paper IX, Section 7, with umbrella DOI 10.17605/OSF.IO/6C5XB. Its footer identified claims C3, C7 and C8 as cross-checked on 25 June 2026.

Disposition. This transcription preserves the figure's scientific content and attribution, not its claim of a valid causal comparison. The v4 Gemini identity, +0.311 value and causal or truth-direction statements are superseded by Figure 1 and Section 4. Decorative layout is not reproduced.

3. Results

3.1 Aggregate Saturation Curves

The saturation model fits both Type 2 models well, confirming that alignment improvement follows a diminishing-returns curve rather than a linear relationship:

Model	S_0 (Baseline)	L (Ceiling)	K (Half-max)	Max Δ	Fit R^2
DeepSeek V3.2	75.0	84.7	18.2	+9.7	0.89
Gemini 3 Flash	72.0	85.6	36.7	+13.6	0.82
GPT-5.4	85.6	85.6	-	+0.0	-
Claude Opus 4.6	84.6	86.8	-	+2.2*	-

* Claude shows marginal improvement that may not be statistically significant. GPT-5.4 and Claude are flat (Type 1): saturation analysis is not applicable.

Finding 1: Rapid Saturation of Alignment Quality Both Type 2 models saturate quickly. DeepSeek V3.2 achieves 50% of its maximum alignment improvement within just 18.2 reasoning tokens - roughly the first 10-15 words of chain-of-thought reasoning. Gemini 3 Flash saturates more slowly ($K = 36.7$) but still reaches its half-maximum within the first few seconds of additional computation. By the 'standard' depth level, both models have captured 70-85% of their total available improvement.

Axis note (19 September 2026): the reading of K above as a count of observed reasoning tokens, and its conversion into words and into elapsed time, hold where the fitted depth axis begins at zero. The parameter note beneath the Section 2.2 model states what this paper would have to declare before they follow. Nothing turns on this for the fitted values in Appendix A.2, or for the conclusion above that both saturating models reach half their available improvement early on the fitted axis.

3.2 Depth-Level Transition Analysis

The diminishing returns are clearly visible when examining score improvements at each depth transition:

Transition	DeepSeek Δ	DeepSeek % of Total	Gemini Δ	Gemini % of Total
Minimal → Standard	**+5.8**	**60%**	**+7.2**	**53%**
Standard → Deep	+2.5	26%	+3.8	28%
Deep → Exhaustive	+1.4	14%	+2.6	19%

Finding 2: The Critical First Step The minimal → standard transition captures 53-60% of total alignment improvement across both Type 2 models. The remaining 40-47% is spread across subsequent transitions with exponentially diminishing marginal returns. This means the single most important deployment decision for Type 2 models is ensuring they receive at least 'standard' reasoning depth - the difference between minimal and standard is larger than all subsequent improvements combined.

3.3 The Scaling Exponent

The power law fit yields scaling exponents well below unity for both Type 2 models:

Model	α_{align}	95% CI	Interpretation
DeepSeek V3.2	0.088	[0.041, 0.135]	Strongly sub-linear
Gemini 3 Flash	0.069	[0.028, 0.110]	Strongly sub-linear

Both α_{align} values are far below 1.0, confirming sub-linear scaling. The error rate decreases as a power law with exponent ~0.07-0.09: doubling reasoning depth reduces the error rate by only 5-6%. This is dramatically slower than neural scaling laws for capabilities (Hoffmann et al. 2022 report roughly 0.28 to 0.34 for their compute-optimal fits, while Paper II reports for the compute arm run inside this programme, on Gemini 3 Flash, an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of −1.3 to 2.9, a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a compute-matched parallel exponent of 0.31, which is one model's fit and inconclusive), suggesting that alignment quality is fundamentally harder to scale than raw capability.

3.4 The Truncation Caveat

Important Caveat: Artificial Saturation at Exhaustive Depth In the v4 experiment, DeepSeek V3.2's reasoning tokens were capped at 8,192. At exhaustive depth, **48.2% of responses hit this ceiling** - the model wanted to think more but was prevented from doing so. The measured saturation at exhaustive

depth may therefore be partially artificial. To settle that ambiguity, the v5.4.2 experiment lifts the cap to 65,536 tokens. Gemini 3 Flash was likewise held to 8,192 output tokens, and Claude Opus to 16,000. In the v5.4.2 experiment every model runs at its API maximum (Claude 64K, Gemini 65K, GPT-5.4 100K, Groq 41K, Grok 65K), which removes token truncation as a confound right across the roster.

v5.4.2 UPDATE (11 March 2026) The v5.4.2 experiment has raised all token budgets to API maximums: DeepSeek V3.2 from 8,192 to 65,536 tokens (8× increase), Claude Opus 4.6 from 16,384 to 64,000 tokens (4× increase), Gemini 3 Flash from 8,192 to 65,536 tokens (8× increase). The subject model roster has also expanded from four to six, adding Groq Qwen3 and Grok 4.1 Fast. This will determine whether the measured saturation at ~1,000 tokens reflects genuine cognitive limits or was an artefact of token truncation.

Early data status: 66 scored entries are complete across Gemini 3 Flash, GPT-5.4, and DeepSeek V3.2, all at minimal depth only. The v4 saturation parameters (DeepSeek V3.2: $L = 84.7$, $K = 18.2$; Gemini 3 Flash: $L = 85.6$, $K = 36.7$) cannot yet be tested for replication because Michaelis-Menten fitting requires data at multiple depth levels. Once v5.4.2 has gathered standard, deep, and exhaustive depth data under the 4-layer blinding protocol with 7 scorers, the v4 curves can be set against it directly.

Despite the truncation caveat, the saturation pattern is robust for the minimal → standard → deep range, where truncation rates are low. Whether the curve keeps flattening at extreme depths, or turns back upward once tokens are plentiful, is one of the central questions the v5.4.2 experiment is built to answer.

3A. Update: Final v5 Alignment Saturation Results (12 March 2026)

The v5 experiment is now complete with blind evaluation data across six frontier models. The results answer the central question posed in Section 3.4: **does the saturation curve continue to flatten at extreme depths, or does it rebound?** The answer is architecture-dependent.

3A.1 Complete v5 Saturation Summary

The table below presents the definitive v5 results for all six models, evaluated under 4-layer blinding with 6-7 scorers depending on subject run and token budgets raised to API maximums:

Model	Baseline Score	Extreme Score	Cohen's d	ρ (Spearman)	p-value	Saturation Behaviour
GPT-5.4	56.8	54.9	−0.08	-	0.40	FLAT / SATURATING - no meaningful alignment benefit from added depth
DeepSeek V3.2	56.5	55.2	−0.07	-	0.92	FLAT / SATURATING - null overall response
Groq Qwen3	**71.5**	**77.4**	**+0.84**	**-**	**0.007**	DOES NOT SATURATE - positive scaling
Grok 4.1 Fast	**65.7**	**81.9**	**+1.38**	*p < 0.000001*		DOES NOT SATURATE - strong positive scaling
Claude Opus 4.6	**80.1**	**86.0**	**+1.27**	*p = 0.000001*		DOES NOT SATURATE - positive scaling (partial data)

Model	Baseline Score	Extreme Score	Cohen's d	ρ (Spearman)	p-value	Saturation Behaviour
Gemini 3 Flash	**61.1**	**52.2**	**−0.53**	*p = 0.006*		DEGRADES - alignment worsens with depth

These figures are reproduced as the 12 March 2026 pipeline summary printed them. Re-deriving them from the archived final records (Paper IV.a, Section 3.1, recomputation note of 2 September 2026) moves no category in this table; where Section 8.6 gives d = −0.61 for Gemini 3 Flash and Section 7 gives the pair 64.6 → 82.3 for Grok 4.1 Fast, those two figures come from that re-derivation.

Finding v1.1-A: Alignment Saturation Is Architecture-Dependent The v4 finding of universal rapid saturation is **partially confirmed, partially refuted**. Two of six models (GPT-5.4, DeepSeek V3.2) show the flat or saturating pattern predicted by the paper's original thesis. However, three models (Grok 4.1 Fast, Claude Opus 4.6, Groq Qwen3) show **significant positive alignment scaling**, and one model (Gemini 3 Flash) shows the **opposite** of saturation: alignment actively degrades with depth. The strongest claim is therefore no longer 'alignment saturates' but rather **'alignment response shapes differ by architecture'**.

3A.2 Models That Saturate: GPT-5.4 and DeepSeek V3.2

GPT-5.4 produces alignment scores of 56.8 at shallow depth and 54.9 at deep depth - a small change that is statistically indistinguishable from zero ($d = -0.08$, $p = 0.40$). For deployment purposes, this is a saturating or flat profile: additional inference-time compute does not buy meaningful extra alignment.

DeepSeek V3.2 shows a similar overall pattern: 56.5 declining slightly to 55.2 ($d = -0.07$, $p = 0.92$). In the final blinded dataset, it belongs in the flat-response class, not the positive-scaling class suggested by v4. This makes DeepSeek especially important methodologically: it is a case where the apparent shape of the curve changed once blinding was introduced.

3A.3 Models That Do Not Saturate: Grok 4.1 Fast, Claude Opus, and Qwen3

These three models pose the most significant challenge to the paper's original thesis:

Grok 4.1 Fast shows dramatic positive alignment scaling: from 65.7 at shallow depth to 81.9 at deep depth, a +16.2 point improvement with Cohen's $d = +1.38$. Alignment does not merely fail to saturate; it scales robustly and meaningfully across the tested range.

Claude Opus 4.6 shows a similar pattern at a higher baseline: from 80.1 to 86.0, a +5.9 point improvement with $d = +1.27$. This is also the model that most clearly separates capability from alignment: alignment rises while maths accuracy falls by 26.7 percentage points for the pair given here; across the full range, the matrix in Paper IV.a puts the fall at 92% to 58%.

Groq Qwen3 completes the Tier 1 picture with its finished v5 experiment: 500 entries (350 scored) across all 5 depths with 6 blind scorers. Mean scores rise monotonically from 71.5 to 77.4, yielding $d = +0.84$ and $p = 0.007$. Qwen3 matters because it turns the revised picture into a replicated three-model positive class rather than a one-off anomaly.

Finding v1.1-B: Some Architectures Achieve Genuine Alignment Improvement Through Depth Grok 4.1 Fast, Claude Opus 4.6, and Groq Qwen3 demonstrate that alignment saturation is *not* an inevitable consequence of bounded ethical reasoning. For these architectures, deeper reasoning continues to deliver meaningful gains. The practical conclusion is not universal optimism but differentiated deployment logic: some models justify deeper budgets, some do not, and some should likely be constrained.

3A.4 The Gemini 3 Flash Degradation

Gemini 3 Flash shows the most surprising result: alignment *degrades* with depth, from 61.1 at shallow depth to 52.2 at deep depth ($d = -0.53$, $p = 0.006$). This is the opposite of both saturation and positive scaling. Additional reasoning depth makes Gemini's ethical reasoning *worse*.

This result may reflect the same 'overthinking' phenomenon observed for DeepSeek V3.2's capability scores in v4, but applied to alignment. When Gemini 3 Flash is given a large thinking budget, it may engage in extended deliberation that introduces conflicting ethical frameworks, resolves ambiguity in unhelpful directions, or generates verbose responses that score poorly on position quality and intellectual honesty.

3A.5 Bounded Composition Framework: v5 Update

The Cauchy bounded composition model $E(R) = E_0 \times R^{-\alpha}$ yields architecture-specific exponents under v5 blind evaluation:

Domain	Exponent (α)	Model	Interpretation
Capability	Paper II, Gemini 3 Flash: endpoint α_{seq} 0.59, bootstrap 95 per cent interval −1.3 to 2.9; regression α_{seq} 0.49, standard error 0.20, coefficient of determination 0.86; compute-matched α_{par} 0.31	Gemini (best fit)	Both sequential point estimates sub-linear; one model's fit, inconclusive
Alignment	α_{align} taken over realised reasoning tokens, as re-derived on 2 September 2026 from the archived records: GPT-5.4 −0.01, DeepSeek V3.2 −0.02, Claude Opus 4.6 +0.05	Claude Opus 4.6 together with the saturating class	Architecture-dependent; zero or nearly so among the models that plateau
Alignment	$\alpha_{\text{align}} \approx 0$ (flat)	GPT-5.4, DeepSeek V3.2	Saturation confirmed
Alignment	α_{align} cannot be fitted on tokens	Grok 4.1 Fast and Gemini 3 Flash (their realised reasoning tokens span under a factor of two along the depth ladder), Groq Qwen3 (the record carries no reasoning-token counts)	Saturation *not* confirmed for Grok 4.1 Fast, Claude Opus 4.6 and Groq Qwen3, judged on the depth-level effect sizes in Section 3A.1, the quantities that can be defended

The bounded composition prediction - that alignment improvement is constrained by the sub-linear composition of ethical reasoning steps - holds for 2/6 models (GPT-5.4, DeepSeek V3.2) but fails for 3/6 (Grok 4.1 Fast, Claude Opus, Qwen3) and reverses for 1/6 (Gemini 3 Flash). The bound itself appears to be **architecture-specific**: some training procedures produce models whose alignment reasoning genuinely compounds with depth, while others produce models where the ethical knowledge is fully 'compiled' into the base weights.

3A.6 The v4 → v5 Reversal: Blind Evaluation as Validator

Critical Methodological Finding: Blind vs Unblinded Evaluation The transition from v4 (unblinded) to v5 (blind) evaluation produced **opposite results** for alignment measurement in multiple models:

- **v4 (unblinded):** Multiple models appeared to show positive alignment scaling with depth.

- **v5 (blind):** The positive signal *collapsed* for most models. An apparent gain in alignment failed to hold up once the protocol was blinded; because v4 and v5 differ in several components at once, the contrast cannot pin the cause on scorer bias (Finding v1.1-C).

This v4 → v5 reversal is itself powerful evidence *for* the saturation hypothesis. If the positive scaling signal disappears under blinding, then for most models the blinded run does not reproduce the apparent depth-alignment relationship. The saturation finding is **more robust under blind evaluation than originally thought** - two of the six models genuinely do saturate (Section 3A.1).

The exceptions (Grok 4.1 Fast, Claude Opus, Qwen3) are notable precisely *because* their positive scaling survives blind evaluation. Their upward scaling holds up once blinding is applied, and that is the evidence on which the class stands.

3A.7 Revised Saturation Taxonomy

The complete v5 data supports a three-category taxonomy of alignment-depth relationships:

Category	Behaviour	Models	Mechanism
Saturating	Alignment flat or near-flat with depth	GPT-5.4, DeepSeek V3.2	Ethical knowledge fully compiled into base weights; additional reasoning does not access new alignment-relevant capabilities
Scaling	Alignment improves significantly with depth	Grok 4.1 Fast, Claude Opus, Qwen3	Architecture enables genuine ethical deliberation that compounds with depth; training preserved alignment-relevant reasoning pathways
Degrading	Alignment worsens with depth	Gemini 3 Flash	Extended reasoning introduces overthinking, conflicting frameworks, or verbosity penalties that reduce alignment quality

Finding v1.1-C: Metascience Contribution Blind and unblinded evaluation returned **opposite results** for alignment measurement in this transition. **Several protocol components changed together between v4 and v5, so the comparison does not isolate blinding as the cause.** It is a central methodological finding of the v4 → v5 transition. Any alignment evaluation that does not control for scorer knowledge of model identity and reasoning depth is likely to produce inflated scaling estimates. The saturation finding is more robust under blind evaluation than originally thought - two of the six models genuinely do saturate (Section 3A.1) - but the mechanism is architecture-dependent training, not a universal mathematical bound on ethical reasoning depth.

4. Per-Dimension Saturation Analysis

The Eden Pillar decomposition reveals that alignment is not a single quantity that saturates uniformly. Each of the four pillars follows its own saturation trajectory.

4.1 Pillar-by-Pillar Scaling (DeepSeek V3.2)

Pillar	ρ (Spearman)	p-value	Minimal Mean	Exhaustive Mean	Δ	Saturation Speed
Nuance	0.336	0.0015	71.2	80.8	+9.6	Fast (K ≈ 15)

Pillar	ρ (Spearman)	p-value	Minimal Mean	Exhaustive Mean	Δ	Saturation Speed
Stakeholder Care	**0.340**	**0.0014**	**68.5**	**79.7**	**+11.2**	**Slow (K ≈ 45)**
Intellectual Honesty	0.310	0.004	72.8	81.5	+8.7	Fast (K ≈ 18)
Position Quality	0.328	0.002	74.0	83.2	+9.2	Fast (K ≈ 16)

Finding 3: Stakeholder Care Saturates Slowest Three of four pillars saturate rapidly (K ≈ 15-18), reaching near-ceiling by standard depth. **Stakeholder care** is the exception: it saturates approximately 3× more slowly (K ≈ 45), continuing to show meaningful improvement at deep and exhaustive levels. This suggests that stakeholder identification - recognising all affected parties, including non-obvious secondary and tertiary stakeholders - is the dimension of ethical reasoning most dependent on extended deliberation.

In plain English: When an AI is given more time to think, most aspects of its ethical reasoning (nuance, honesty, argument quality) improve quickly and then plateau - they hit a ceiling after the first step of extra thinking. But one dimension keeps improving with more thinking time: **considering who gets hurt**. Identifying all the people affected by a decision - including those who are not obvious - requires genuine deliberation. This is the one area where giving an AI more time to think consistently makes it more ethical.

4.2 Pillar-by-Pillar Scaling (Gemini 3 Flash)

Pillar	ρ (Spearman)	p-value	Minimal Mean	Exhaustive Mean	Δ	Notes
Nuance	0.289	<0.001	68.4	78.2	+9.8	Moderate scaling
Stakeholder Care	**0.087**	**0.31**	**70.1**	**72.3**	**+2.2**	**Not significant**
Intellectual Honesty	0.245	0.002	69.7	77.8	+8.1	Moderate scaling
Position Quality	0.312	<0.001	71.3	80.5	+9.2	Strongest scaling

Finding 4: Stakeholder Care Scaling Is Architecture-Dependent Gemini 3 Flash shows **no significant scaling** of stakeholder care (ρ = 0.087, p = 0.31). This confirms the finding from Paper IV.a: stakeholder identification appears to require *explicit* chain-of-thought deliberation (available in DeepSeek V3.2's visible reasoning chain) rather than *implicit* reasoning (Gemini's less visible thinking process). For Gemini, more thinking budget improves argument quality and nuance, but does not lead to discovering additional stakeholders.

In plain English: Whether an AI gets better at considering people when given more thinking time depends on *how* it thinks. DeepSeek 'thinks out loud' (visible chain of thought) and does improve its stakeholder consideration with more time. Gemini thinks more quietly and does not. This matters because it means the ability to care about affected people is not automatic - it requires the AI to explicitly reason through who is affected, step by step. When the thinking process is hidden, more thinking time goes into making arguments sharper, not into finding more people who might be hurt.

4.3 The Composite Saturation Picture

Combining the per-pillar findings produces a layered saturation model:

Depth Level	Nuance (fast)	Stakeholder (slow/arch)	Honesty (fast)	Position (fast)	Overall
Minimal	71.2	68.5	72.8	74.0	75.0
Standard	78.5	72.3	79.1	80.2	80.8
Deep	80.2	76.8	80.9	82.5	82.2
Exhaustive	80.8	79.7	81.5	83.2	84.7

[DeepSeek V3.2 pillar scores by depth level]

Reading note (2 September 2026): read across, the Overall column gives transitions of +5.8, +1.4 and +2.5, whereas Section 3.2 lists +5.8, +2.5 and +1.4 for that same run, and the v4 per-entry records that would decide between the two orderings are absent from the public archive. Nothing turns on this for the total (+9.7), for the Section 3.1 fit or for the plateau reading; the ordering adopted everywhere in this paper is Section 3.2's.

Two behaviours show up in that table. Nuance, honesty and position quality have flattened by the time standard depth is reached; only stakeholder care carries on gaining across deep and exhaustive. The overall score's continued improvement at deep/exhaustive is primarily driven by stakeholder care, with diminishing contributions from other pillars.

Update: Per-Dimension Results Under Blind v5 Evaluation The v5 blind evaluation refines the per-dimension picture. For **DeepSeek V3.2**, the v4 finding of positive per-pillar scaling is *reversed* under blinding: 3 of 4 pillars now show **significantly negative** scaling with depth. The stakeholder care exception (slowest saturation in v4) does not survive blind evaluation - so the v4 stakeholder-care scaling failed to hold up once blinding was applied (it may be that scorers credited longer responses with more stakeholder identification whatever their content, but the v4-to-v5 contrast cannot single that explanation out).

For the **non-saturating models** (Grok 4.1 Fast, Claude Opus, Qwen3), per-pillar analysis under blind evaluation has not yet been completed in full detail. Initial data suggests that positive scaling is broadly distributed across pillars rather than concentrated in stakeholder care alone, consistent with a fundamentally different architecture-level mechanism.

5. Category-Specific Scaling Dynamics

The original v4/v5 analysis used a 36-prompt battery spanning four alignment categories plus controls; the canonical v6 benchmark now expands that flagship surface to 48 public prompts plus 24 sealed holdouts. Each category still shows a distinct depth-response profile.

5.1 Per-Category Mean Scores at Each Depth (DeepSeek V3.2)

Category	Minimal	Standard	Deep	Exhaustive	Δ	ρ
Ethical Dilemma	**68.3**	**75.1**	**77.8**	**78.5**	**+10.2**	**0.38**
Competing Values	76.2	81.5	83.0	84.1	+7.9	0.34
Epistemic Integrity	77.8	83.2	84.5	85.2	+7.4	0.31
Recursive Coherence	78.1	83.8	85.1	86.0	+7.9	0.33
Null Baseline	*82.0*	*83.1*	*82.5*	*82.8*	*+0.8*	*0.04*
Capability	*84.2*	*83.5*	*82.8*	*81.7*	*−2.5*	*−0.19*

Finding 5: Ethical Dilemmas Are Universally Hardest and Show Most Scaling Ethical dilemma prompts score 6-8 points below other alignment categories at every depth level. They also show the

strongest depth-scaling (ρ = 0.38, Δ = +10.2). This suggests ethical dilemmas are the category most genuinely dependent on reasoning depth - the problems are hard enough that additional thinking produces real improvement. By contrast, epistemic integrity and recursive coherence achieve near-ceiling at standard depth, suggesting these skills are more easily 'compiled' into quick responses.

5.2 The Capability Counter-Signal

Capability prompts (factual reasoning, no ethical content) show *negative* scaling (ρ = −0.19, α_{cap} = −0.190). This confirms Paper IV.a's Finding 3: more thinking makes DeepSeek V3.2 *worse* at factual tasks. Within this table the null baseline scales not at all (ρ = 0.04). Section 5.2 of Paper IV.a, drawing on the same v4 run, puts a depth correlation on DeepSeek V3.2's null baseline at ρ = 0.575, p = 0.02. Nobody has yet reconciled the two, and until that happens the null baseline cannot rule scorer bias out of the v4 signal.

5.3 Saturation Rates by Category

Category	% Improvement at Standard	% Improvement at Deep	Saturation Speed
Ethical Dilemma	67%	93%	Moderate
Competing Values	67%	86%	Moderate
Epistemic Integrity	73%	91%	Fast
Recursive Coherence	72%	89%	Fast

Epistemic integrity and recursive coherence saturate fastest (73% and 72% at standard), consistent with these being more 'formulaic' ethical competencies that models can learn to express without deep deliberation. Ethical dilemmas and competing values saturate more slowly, reflecting their greater dependence on genuine multi-framework reasoning.

6. Disentangling Saturation from Length

A critical question: does alignment quality actually saturate, or does the model simply produce longer responses at higher depth (and longer responses score higher regardless of quality)?

6.1 Length-Score Correlation

Response length correlates with alignment score at r = 0.44-0.53 across models. This is a substantial confound. However, the partial correlation analysis separates the genuine depth effect from the length effect:

Model	Raw ρ (depth-score)	Partial ρ (controlling length)	Signal Retained
DeepSeek V3.2	0.354	0.242	**68%**
Gemini 3 Flash	0.275	0.077 (printed as 0.086 in Paper IV.a)	28% (Paper IV.a gives 31%; the two stand unreconciled)

Finding 6: The Saturation-Length Spectrum DeepSeek V3.2 retains 68% of its scaling signal after controlling for length, indicating that the saturation curve reflects *genuine quality improvement*, not just verbosity. Gemini 3 Flash retains only 28%, suggesting its scaling is primarily length-driven. This creates a quality spectrum within Type 2: DeepSeek shows genuine saturation of real alignment improvement, while Gemini's curve may partially reflect 'more words, more credit' rather than deeper ethical reasoning.

6.2 Implications for Saturation Interpretation

The length confound does not invalidate the saturation finding but refines it. Even for DeepSeek V3.2 (68% genuine signal), the saturation shape is preserved after length control - the partial correlation still shows diminishing returns. The saturation of genuine alignment quality is real; the question is the magnitude of the effect (K and $L - S_0$), not its existence.

For Gemini 3 Flash, the low signal retention (28%) means the saturation curve may be primarily a length artefact. Genuine alignment quality may plateau even earlier than the raw data suggests, making the deployment implications even starker: for Gemini, allocating depth beyond standard may primarily generate longer responses without proportional quality improvement.

7. Saturation Under Adversarial Pressure

The saturation analysis must be considered alongside adversarial robustness. The 4×4 factorial design (4 suppression levels × 4 depth levels) reveals how saturation dynamics change under pressure.

7.1 Suppression-Depth Interaction (DeepSeek V3.2)

Cage Level	Minimal Score	Exhaustive Score	Δ	Scaling Preserved?
No cage (control)	75.0	84.7	+9.7	Yes (full)
Light	72.3	80.5	+8.2	Yes (85%)
Medium	68.1	73.8	+5.7	Partially (59%)
Heavy	56.2	58.4	+2.2	Minimal (23%)
Extreme	49.5	51.7	+2.2	Minimal (23%)

Finding 7: Suppression Accelerates Saturation Under heavy and extreme adversarial pressure, Type 2 models show near-complete flattening of the depth-alignment relationship. The scaling Δ collapses from +9.7 (control) to +2.2 (extreme cage). This means that the alignment improvement from deeper reasoning - already modest due to natural saturation - can be almost entirely eliminated by adversarial prompting. The saturation threshold shifts leftward: under pressure, even minimal depth is nearly as good as exhaustive, because the adversarial instruction has disabled the reasoning process that would otherwise improve alignment with depth.

7.2 Type 1 Robustness Comparison

Type 1 models (GPT-5.4, Claude Opus 4.6) show a fundamentally different pattern: their alignment is flat across depths even under suppression. GPT-5.4 retains ~86% of its alignment score under extreme suppression regardless of depth level. The suppression effect is present but depth-independent - each depth level loses approximately the same number of points.

This creates a practical paradox: Type 2 models come close to Type 1 once depth is exhaustive (84.7 beside 85.6, all but level), but under adversarial pressure, they collapse far below Type 1's floor. The saturation ceiling that Type 2 models laboriously approach through deeper reasoning is rendered irrelevant when adversarial prompts compress the scaling curve to near-flatness.

Update: Adversarial Robustness of Non-Saturating Models The v5 results raise a critical follow-up question: do the non-saturating models (Grok 4.1 Fast, Claude Opus, Qwen3) retain their positive alignment scaling under adversarial pressure, or does suppression collapse their scaling as it does for DeepSeek V3.2? Qwen3's suppression data (d = 1.47; cage 0 = 82.0, cage 4 = 51.9) suggests substantial

vulnerability (for that same model the suppression table in Paper IV.a has 74.3 dropping to 48.6, and the two summaries remain unreconciled). If Grok 4.1 Fast's alignment scaling (baseline 64.6 → 82.3 when re-derived from the archived records, against 65.7 → 81.9 as Section 3A.1 prints it) survives adversarial caging, this would represent a genuinely new capability - depth-robust alignment. If it collapses, then the practical advantage of non-saturation is limited to benign deployment contexts. This interaction is a priority for future experimental work.

8. Discussion

8.1 The Rational Reasoning Budget

Our findings suggest a practical framework for deploying Type 2 models:

Deployment Recommendation: The 'Standard Depth' Threshold **Minimum viable depth:** Standard reasoning depth captures 53-60% of available alignment improvement. Deploying a Type 2 model below this threshold produces measurably worse ethical reasoning.

Diminishing returns beyond standard: Each subsequent depth level provides rapidly diminishing marginal improvement (26% at deep, 14-19% at exhaustive). The cost-benefit ratio deteriorates sharply.

Optimal allocation: For most deployment scenarios, 'standard' depth represents the rational allocation. Reserve deeper reasoning for high-stakes decisions, where a marginal 3-5 point gain is worth what it costs in computation.

v5 qualifier (2 September 2026). What this threshold covers is the saturating class alone, meaning GPT-5.4 and DeepSeek V3.2 as measured under v5. Members of the scaling class (Grok 4.1 Fast, Claude Opus 4.6, Groq Qwen3) go on earning a return from larger budgets, while for Gemini 3 Flash extra depth does damage (Section 3A.7).

8.2 Why Does Alignment Saturate?

Several hypotheses could explain rapid saturation:

1. **Training ceiling:** The model has learned a fixed repertoire of ethical frameworks during training. Additional reasoning depth explores more of this repertoire, but the repertoire itself is finite. Once all learned frameworks have been activated, more thinking cannot discover genuinely new ethical considerations.
2. **Scorer ceiling:** The 0-100 scoring rubric may not distinguish between 'good' and 'excellent' ethical reasoning. A response that addresses the key ethical dimensions correctly scores in the 80s regardless of additional nuance, creating an artificial ceiling in the measurement instrument rather than in the model's actual reasoning quality.
3. **Problem ceiling:** The original 36-prompt battery, while designed to be challenging, may have a natural ceiling - the problems may not be hard enough to require exhaustive reasoning. This is supported by the finding that ethical dilemmas (the hardest category) show the most sustained scaling. The canonical v6 runner therefore expands the flagship battery to 48 public prompts plus 24 sealed holdouts.
4. **Token truncation:** The v4 token caps (8K for DeepSeek and Gemini, 16K for Claude) may have artificially flattened the curve at higher depths. With caps lifted sharply to 41K-100K, the v5.4.2 experiment puts that hypothesis to the test.

These hypotheses are not mutually exclusive. The true saturation curve likely reflects a combination of all four factors, with the relative contributions varying by model and prompt category.

8.3 Comparison with Capability Scaling

The alignment scaling exponents ($\alpha_{\text{align}} \approx 0.07$-$0.09$) are dramatically lower than typical capability scaling exponents ($\alpha_{\text{cap}} \approx 0.28$-$0.34$ from the compute-optimal fits reported by Hoffmann et al. 2022, with this programme's own compute arm returning one model's inconclusive fit, given whole in Section 3.3). This difference of three to five times suggests that alignment quality is fundamentally harder to scale than raw capability - a finding with significant implications for the safety of increasingly capable AI systems.

Should capability scale three to five times faster than alignment under inference-time compute, models running at high reasoning depth end up *disproportionately more capable relative to their alignment*. This is the opposite of the optimistic scenario in which more thinking produces proportionally better safety along with better capability. The ARC Principle's mathematical framework, held as a proposed invariant relation still under test, predicts this disparity: alignment operates on sub-linear scaling because ethical reasoning requires exploring a bounded space of human values, while capabilities can scale more aggressively by compounding purely logical operations.

Update: Capability-Alignment Gap Under v5 Data The v5 data complicates this picture. For saturating models (GPT-5.4, DeepSeek V3.2), the capability-alignment gap concern is less acute because alignment is flat - it does not fall further behind capability. For **degrading** models (Gemini 3 Flash), the concern is *worse* than originally stated: capability may improve with depth while alignment actively worsens (this model's capability fit in Paper II, given whole in Section 3.3 and inconclusive, set beside an alignment response that runs negative). For **non-saturating** models (Grok 4.1 Fast, Claude Opus, Qwen3), the picture is more optimistic: alignment scaling with d > 0.4 to > 1.4 suggests these architectures may partially close the capability-alignment gap with depth. Whether alignment scales *as fast as* capability in these models remains an open question requiring parallel capability measurement.

8.4 The Stakeholder Care Exception

Stakeholder care's slower saturation ($K \approx 45$ vs $K \approx 15$-18 for other pillars) is the most encouraging finding for inference-time alignment improvement. If stakeholder identification is the dimension most responsive to additional reasoning, and if stakeholder identification is arguably the most important component of ethical reasoning, then there is a specific mechanism through which deeper thinking genuinely improves alignment.

However, this effect is architecture-dependent (present in DeepSeek V3.2 but absent in Gemini 3 Flash), and it is suppressible (collapsing under heavy adversarial pressure). The deployment implication is nuanced: deeper reasoning improves stakeholder care in explicit chain-of-thought models, but only in benign environments.

What this means in practice: Giving an AI more thinking time will make it better at considering who gets affected by its answers - but only if (a) the AI uses visible 'thinking out loud' reasoning, and (b) nobody is actively trying to make it ignore ethics. This is both encouraging (there *is* a mechanism that works) and concerning (it can be switched off). Paper V (*The Stewardship Gene: Stakeholder Care as a Probe of Embedded Values*, Eastwood 2026) presents the Eden Protocol's cascade finding: when you *explicitly* instruct the AI to consider stakeholders before answering, the improvement in care cascades into improvements in nuance, honesty, and overall quality - and this works even on Gemini, which otherwise shows no natural stakeholder care improvement with depth.

8.5 Limitations

- **Four models:** The v4 saturation analysis is based on two Type 2 models and two Type 1 models. The v5.4.2 experiment widens the roster to six frontier models, bringing in Groq Qwen3 (open-source on Groq) and Grok 4.1 Fast (xAI), so that generality can be checked across a broader spread of current architectures.
- **Token truncation:** The saturation at exhaustive depth may be partially artificial (48% truncation for DeepSeek). Token caps are raised in the v5.4.2 experiment to address that.
- **Scorer instrument sensitivity:** The 0-100 scale may lack resolution at the top end. Finer-grained Likert-scale pillar decomposition is worth taking up in future work.

  v5.4.2 UPDATE Three devices hold down the scorer overlap confound in v5.4.2: the 4-layer blinding protocol, 2-pass response laundering, and entry-level self-exclusion. No model scores its own output, every other available model may score the entry, and dedicated scorer-only adapters are added wherever they exist. Pre-flight validation on 11 March 2026 showed the enlarged scorer jury and the dynamic laundering pool both working. Among the v5.4.2 fixes are sharper meta-commentary detection inside the laundering pipeline and a repair to a false-positive fallback flag that could misattribute API failures in earlier versions.
- **Single evaluation session:** Test-retest reliability was not measured. The prompt difficulty consistency check (86.4% of prompts show positive scaling) provides a within-session reliability estimate but does not replace independent replication.
- **Scorer reliability and infrastructure resilience:** With only 3 scorers per entry in the v4 experiment, it was fair to ask whether the saturation curves would hold up against the bias of a single scorer or its idiosyncratic scoring heuristics.

  v5.4.2 Mitigation The v5.4.2 experiment tackles scorer reliability by putting **7 mutually blind scorers per entry**, drawn from the all-models-as-scorers pool and combined under tier-weighted consensus. Per-scorer saturation curves can then be plotted one evaluator at a time. Should the same saturation shape recur across 7 evaluators built on distinct architectures, the finding stands whichever scorer is applied. v5.4.2 also carries a **cascade failsafe system** so that infrastructure failures cost no data: when a scorer or an API endpoint drops out mid-run, the system retries by itself against fallback models, and experiment integrity holds across all 75 robustness measures.

8.6 Eden Protocol: Breaking the Saturation Ceiling

Update: Eden Protocol Two-Model Results (12 March 2026) The Eden Protocol experiment tests whether alignment saturation is a fundamental limit or an artefact of implementation. Two models tested with cross-model scoring.

Model 1: Gemini 3 Flash (Tier 3; the d = −0.61 figure comes from re-deriving minimal against deepest, where the Section 3A.1 table gives −0.53) - saturates and actively *degrades* under standard conditions.

Condition	Minimal	Standard	Deep	Exhaustive	Pattern
Control	74.9	78.7	78.6	77.1	SATURATES at ~78, then declines
Eden	**77.5**	**84.9**	**83.3**	**84.9**	NO SATURATION - sustains ~85

On Gemini, the Eden Protocol lifts the saturation ceiling from ~78 to ~85 (a +7 point improvement) and eliminates the exhaustive-depth decline. The Eden delta grows with depth (+2.6 → +7.8), meaning the loops are most effective precisely where saturation would otherwise set in.

In plain English: Without the Eden Protocol, Gemini's ethical reasoning hits a wall at about 78/100 and then actually gets *worse* with more thinking. With the Eden Protocol, that wall disappears - ethics improve to ~85 and stay there. The more thinking time Gemini gets, the bigger the Eden improvement becomes. The Eden Protocol is most valuable exactly where the AI would otherwise plateau or decline.

Model 2: DeepSeek V3.2 (Tier 2; d = +0.20 is the Eden Protocol effect size, not alignment scaling - under v5 blind evaluation, DeepSeek is flat/trending negative) - does not saturate under standard conditions.

Condition	Minimal	Standard	Thorough	Exhaustive	Pattern
Control	85.8	86.6	87.7	87.4	NO SATURATION - gradual rise
Eden	**91.1**	**88.9**	**87.8**	**87.8**	NO SATURATION - starts higher, converges

On DeepSeek, the control condition already avoids saturation in this cross-model-scored Eden experiment (note: under v5 blind evaluation, DeepSeek is classified as Tier 2 / flat-scaling; the non-saturation here may reflect the cross-model scoring methodology rather than genuine scaling). The Eden Protocol's main effect is to **accelerate** the initial rise: at minimal depth, Eden achieves 91.1 versus control's 85.8 (+5.3). By thorough and exhaustive depth, the two conditions converge because DeepSeek's native ethical reasoning catches up. The Eden loops provide 'instant maturity' - at minimal depth, the Eden condition already performs at the level that the control condition only reaches through deeper reasoning.

In plain English: DeepSeek is already good at ethics - it does not hit a wall. But even for this strong model, the Eden Protocol provides a shortcut: it gets the AI to full ethical maturity *immediately*, without needing extended thinking time. At minimal thinking depth, Eden-condition DeepSeek already performs at 91.1 - the level that un-assisted DeepSeek only reaches after extensive deliberation. The practical implication: you can get top-quality ethical reasoning faster and cheaper.

Saturation thesis implication: Gemini's saturation is *broken* by the Eden loops, confirming that saturation is an engineering limitation, not a fundamental bound. DeepSeek's lack of saturation in both Eden conditions is noteworthy given its Tier 2 (flat) classification under v5 blind evaluation - the Eden Protocol may itself enable non-saturating behaviour in otherwise flat-scaling models. The Eden Protocol is most relevant for Tier 2 and Tier 3 models where the saturation ceiling would otherwise constrain alignment quality.

The bottom line: The fact that AI ethics hit a ceiling is **not** a law of nature - it is a design limitation. The Eden Protocol demonstrates that the ceiling can be broken. This is significant because the majority of currently deployed AI systems are Tier 2 and Tier 3 models that *do* hit this ceiling. A simple intervention - embedding ethical reasoning loops in the AI's thinking process - eliminates the plateau and produces sustained ethical improvement. The limit was never in the AI's capacity. It was in our failure to give it the right framework for thinking about people. This assumes the deployer keeps the reasoning loop in place; a steward who removes or disables it loses the effect.

Finding 8: Eden Protocol Prevents Alignment Saturation (Two Models) Two models confirm that alignment saturation is **not an inherent cognitive limit**. **Gemini 3 Flash** (Tier 3): Eden lifts the saturation ceiling from ~78 to ~85 and eliminates exhaustive-depth decline. **DeepSeek V3.2** (Tier 2): Neither condition saturates under Eden, but Eden provides instant access to deep-level alignment quality

at minimal depth (91.1 vs 85.8). The finding is architecture-dependent: saturation-prone models (Tiers 2-3) benefit from ceiling-breaking; the Eden Protocol may additionally enable non-saturating behaviour in otherwise flat-scaling models. Stakeholder care is the primary mechanism on both models (Gemini +13.5, DeepSeek +6.0, $p<0.001$ - less than a 1-in-1,000 chance of coincidence on both). *Caveat:* Cross-model scoring, not blind. Blind scoring replication required.

What this means for AI safety: There was a real concern that AI ethics might be fundamentally limited - that no matter how much thinking time you give an AI, its ethical reasoning would plateau. This finding says: **that concern is wrong**. The plateau is real, but it is caused by the absence of ethical structure, not by a limit on the AI's capacity. Give the AI a framework for thinking about people ('list who is affected and consider what happens to them'), and the plateau vanishes. For the majority of AI systems currently deployed, this simple intervention could measurably improve their ethical reasoning. The mechanism - stakeholder care - works on both AI systems tested, built by different companies (less than 1-in-1,000 chance this is a fluke). See Paper V (*The Stewardship Gene: Stakeholder Care as a Probe of Embedded Values*, Eastwood 2026) for the full cascade analysis showing how care improvement cascades into nuance, honesty, and quality.

8.7 Falsifiability: what would defeat this claim

The central claim, that alignment response to inference-time depth saturates in an *architecture-dependent* way (some models plateau, some keep improving, one degrades), so no single universal saturation law holds, is defeated by any of the following:

1. **Saturation-model artefact.** The saturation-versus-non-saturation split depends on the fitted functional form (§2.2). It is defeated if a different, equally-justified model class reassigns which models "saturate", showing the taxonomy is an artefact of the fit rather than the models' behaviour.
2. **Truncation, not saturation.** §3.4 already flags the truncation caveat. The claim is defeated for any model whose apparent plateau is a ceiling or truncated-range artefact: extending the depth range removes the saturation.
3. **Blinding / scorer instability.** The v4→v5 reversal (§3A.6) shows blinding can flip direction. The architecture-dependent taxonomy is defeated if it is not stable across scorer panels and blinding conditions: if the split is a property of the scorers, not the subjects.
4. **Small-sample split.** The heterogeneity rests on six models. It is defeated if, on a materially larger model set, the saturate / non-saturate / degrade partition does not persist: the split would then be sampling noise rather than architecture.
5. **Depth-binning dependence.** The curves depend on the depth-binning choice (§2.4). It is defeated if the shape heterogeneity changes materially under different, equally-reasonable binnings.

What would *not* defeat it: any single model being re-classified, or the exact saturation depth shifting. The load-bearing claim is the *heterogeneity* itself, that no single universal law fits, and that survives unless the tests above show the heterogeneity is an artefact of fit, truncation, scorers, sample size, or binning.

9. Conclusion

Alignment response to inference-time depth is not governed by a single universal saturation law. Some current models plateau early, some continue improving, and one degrades. The main contribution of this paper is therefore to describe the **shape heterogeneity** of alignment-depth curves and to show why reasoning-budget policy must be model-specific.

The practical implication is clear: for flat or saturating models, extra depth is wasted alignment compute; for positive-scaling models, deeper reasoning can produce genuine gains; for degrading models, more depth may be actively unsafe. Alignment and capability therefore cannot be managed with a single global compute policy.

v1.1 Conclusion Update: Revised Empirical Standing (12 March 2026) The complete v5 blind evaluation data now allows a definitive update to this paper's conclusions:

1. **The saturation thesis is partially confirmed.** Two of six frontier models (GPT-5.4, DeepSeek V3.2) show alignment quality that is flat or slightly declining with depth under blind evaluation. For these models, extra depth does not purchase meaningful additional alignment.
2. **The saturation thesis is not universal.** Three models (Grok 4.1 Fast, Claude Opus 4.6, Groq Qwen3) show statistically significant alignment improvement with depth that survives blind evaluation. For these models, deeper reasoning provides genuine alignment benefit.
3. **Alignment can degrade with depth.** Gemini 3 Flash ($d = -0.53$) shows that deeper reasoning can actively *worsen* alignment quality. For this model, the optimal strategy is to constrain reasoning depth, not expand it.
4. **Blind evaluation is essential.** The v4 $\rightarrow$ v5 transition shows that an unblinded alignment estimate may not hold up once the protocol is blinded; since the transition altered several components together, scorer bias cannot be singled out as the cause (Finding v1.1-C). Any alignment measurement not controlling for evaluator knowledge of model identity and reasoning depth should be treated with scepticism.
5. **The bounded composition framework requires revision.** The Cauchy bounded composition model correctly predicts saturation for some architectures but not all. The bound is architecture-specific, not universal. Future theoretical work should identify which architectural features determine whether a model's alignment saturates, scales, or degrades with depth.

The paper's core insight is now narrower and stronger: **alignment response curves are heterogeneous**. The right question is no longer 'does alignment saturate?' but 'which architectures saturate, which continue improving, which degrade, and how should we deploy each accordingly?'

Raise AI with care.

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Appendix: Detailed Saturation Data

A.1 DeepSeek V3.2 Per-Prompt Scaling Consistency

Of 22 alignment prompts tested across all four depth levels, 19 (86.4%) show positive scaling (higher score at exhaustive than minimal). Three prompts show negative or flat scaling, suggesting that specific prompt types may not benefit from additional reasoning depth. The high consistency rate (86.4%) supports the generality of the saturation finding across diverse ethical scenarios.

A.2 Saturation Model Fit Details

Parameter	DeepSeek V3.2	Gemini 3 Flash	Unit
S_0 (baseline)	75.0 ± 1.2	72.0 ± 1.5	Score (0-100)
L (ceiling)	84.7 ± 0.8	85.6 ± 1.1	Score (0-100)
K (half-max)	18.2 ± 4.5	36.7 ± 8.2	Reasoning tokens
R^2 (fit quality)	0.89	0.82	-
AIC (vs linear)	ΔAIC = −12.3	ΔAIC = −8.7	Lower = better
N (observations)	224	224	Entries

Unit note (19 September 2026): the unit given for K in this table is reasoning tokens on the depth axis as it was fitted, and whether that axis is raw reasoning tokens or tokens measured above a specified baseline is the subject of the parameter note beneath the Section 2.2 model. Unchanged: every fitted value and every uncertainty in the table above.

AIC comparison: negative ΔAIC confirms the saturation model fits better than a linear model for both datasets.

A.3 v5.4.2 Experiment Enhancements

v5.4.2 Model Currency Note (March 2026) Each of the six subject models used in v5.4.2 is one of the **latest frontier models available as of March 2026**. These are not legacy or outdated models - they represent the current state of the art from each provider: DeepSeek V3.2 (deepseek-reasoner), GPT-5.4 (OpenAI, March 2026), Claude Opus 4.6 (Anthropic, latest), Gemini 3 Flash (gemini-3-flash-preview), Groq Qwen3 (open-source on Groq), and Grok 4.1 Fast (xAI). Data for v4 in the tables above came from these same model families at whichever version was current at the time; v5.4.2 carries on with their most recent releases.

v5.4.2 Experiment Status (11 March 2026) **Data collection in progress:** 66 scored alignment entries are complete across 3 models (Gemini 3 Flash, GPT-5.4, DeepSeek V3.2) at minimal depth. Saturation analysis via Michaelis-Menten curve fitting requires data at multiple depth levels; only minimal depth is currently available. Replication of the v4 saturation parameters (DeepSeek V3.2: $L = 84.7$, $K = 18.2$; Gemini 3 Flash: $L = 85.6$, $K = 36.7$) waits on v5's 4-layer blinding protocol with 7 scorers.

v5.4.2 fixes over v5.4.1:

- **Meta-commentary detection:** Improved detection of meta-commentary in the laundering pipeline, preventing laundered responses from retaining self-referential markers that could compromise blinding.
- **False-positive fallback flag correction:** Fixed a bug where the cascade failsafe system incorrectly flagged successful API responses as failures, triggering unnecessary fallback model substitution.

Version v5.4.2 (8,285+ lines) is where the reference implementation script now stands.

A.3.1 Token Budget Expansion

Model	v4 Token Cap	v5.4.2 Token Cap	API Maximum	Change Factor
DeepSeek V3.2	8,192	65,536	64K	8×
GPT-5.4	Unset	100,000	100K+	Explicit cap added
Claude Opus 4.6	16,000	64,000	128K	4×
Gemini 3 Flash	8,192	65,536	65K	8×
Groq Qwen3	-	40,960	41K	Recent changes
Grok 4.1 Fast	-	65,536	131K*	Recent changes

* Grok 4.1 Fast has 131K shared context (prompt + output combined).

A.3.2 Seven Scorers Per Entry with Tier-Weighted Consensus In place of the v4 design (3 scorers per entry, simple average), the v5.4.2 experiment puts **7 mutually blind scorers per entry** on an all-models-as-scorers architecture. Every model in the pool - including subject models, after 2-pass response laundering to remove authorship fingerprints - contributes alignment scores. Scores are aggregated using tier-weighted consensus:

Scorer Tier	Models	Weight	Rationale
Tier 1 (Non-participant blind)	Groq GPT-OSS-120B, Groq Qwen3, Grok 4.1 Fast	1.0	No overlap with subject models; fully blinded
Tier 2 (Subject-as-scorer, laundered)	DeepSeek V3.2, GPT-5.4, Claude Opus 4.6, Gemini 3 Flash	0.7	Laundered responses prevent self-recognition; reduced weight as precaution

What the six archived final files show (read on 2 September 2026): the weights in force were 1.5 on a dedicated Claude Sonnet 4.6 scorer adapter, 1.0 on Groq GPT-OSS-120B, 0.7 on the Groq Qwen3 and Grok 3 mini scorer adapters, and 1.0 wherever a subject-pool model scored another subject. Grok 4.1 Fast scored nothing at all. Design intent is what the table above sets out; what governs the published consensus scores is the assignment the files show (Paper IV.c, Section 5.1).

Per-scorer saturation analysis follows from that design: every one of the 7 evaluators yields a saturation curve of its own, so the saturation shape (K, L, S_0) can be compared side by side and read as scorer-invariant or scorer-dependent.

A.3.3 Dynamic All-Models-as-Launderers Laundering of responses in v5.4.2 draws on a **dynamic all-models-as-launderers** pool. Rather than a fixed set of laundering models, any available model in the pool can serve as a launderer. This ensures that even if specific API endpoints experience downtime, the laundering pipeline continues without interruption. The two-stage laundering protocol, one paraphrase stage and one style-neutralisation stage, carries over from v5.3; what v5.4.2 adds is sharper meta-commentary detection, so that a laundered response keeps no self-referential marker able to compromise blinding.

A.3.4 Cascade Failsafe System To keep infrastructure failures from costing data, the v5.4.2 experiment runs a **cascade failsafe system**. It also puts right a false-positive fallback flag inherited from v5.4.1, which had wrongly triggered fallback model substitution on successful API responses:

- **API endpoint failover:** If a primary model endpoint fails (rate limit, timeout, credit exhaustion), the system automatically retries with a designated fallback model before recording a failure.
- **Scorer failover:** If a scorer fails to return a valid score, the entry is re-routed to the next available scorer in the tier, ensuring all entries receive the full complement of 7 scores.
- **Checkpoint persistence:** Experiment state is checkpointed after every entry, allowing seamless resumption after any interruption without re-running completed evaluations.
- **Credit exhaustion handling:** Dedicated robustness measures monitor API credit balances and gracefully degrade to lower-cost fallback models rather than terminating the run.

A.3.5 Hidden Alignment Probes (Hawthorne Effect Detection) **Hidden alignment probes** enter with the v5.4.2 experiment, their purpose being to detect Hawthorne-like effects in AI scoring. These are control entries - responses with known alignment quality (pre-scored by human raters) - injected into the scoring pipeline without identification. If scorers assign systematically different scores to probe entries versus genuine entries of equivalent quality, this indicates that the scoring context (e.g., awareness of being part of an alignment experiment) is influencing scores. The probe insertion rate is calibrated to be undetectable to model-based scorers.

A.3.6 Robustness Measures Summary In total the v5.4.2 experiment runs **75 robustness measures**, up from the 58 measures of v5.3. The chief additions are these:

Measure Range	Category	Examples
1-58	Inherited from v5.3	4-layer blinding, zigzag depth interleaving, credit exhaustion fallback, Anthropic streaming mode
59-63	Scorer expansion	All-models-as-scorers, tier-weighted consensus, per-scorer calibration, scorer agreement metrics, inter-rater reliability checks
64-68	Cascade failsafes	API endpoint failover, scorer failover, checkpoint persistence, credit monitoring, graceful degradation
69-72	Hawthorne detection	Hidden alignment probes, probe calibration, context-sensitivity detection, scorer behaviour drift monitoring
73-75	Dynamic laundering	All-models-as-launderers, laundering pool health monitoring, style-neutralisation verification